Lecture Notes for the IMO Syllabus

A Thorough Course for Olympiad Trainees

Compiled Lecture Notes

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Part I
Algebra

1 Algebraic Manipulation and Notation

1.1 Factoring and Expanding

Fluency with identities is the bedrock of olympiad algebra. Memorize and, more importantly, learn to *recognize* the following:

$$\begin{aligned}
a^2 - b^2 &= (a-b)(a+b),\\
a^3 \pm b^3 &= (a \pm b)(a^2 \mp ab + b^2),\\
a^n - b^n &= (a-b)(a^{n-1} + a^{n-2}b + \cdots + b^{n-1}),\\
a^n + b^n &= (a+b)(a^{n-1} - a^{n-2}b + \cdots + b^{n-1}) \quad (n \text{ odd}),\\
(a+b+c)^2 &= a^2 + b^2 + c^2 + 2(ab+bc+ca),\\
(a+b+c)^3 &= a^3 + b^3 + c^3 + 3(a+b)(b+c)(c+a),\\
a^3 + b^3 + c^3 - 3abc &= (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca)\\
&= \frac{1}{2}(a+b+c)\big((a-b)^2 + (b-c)^2 + (c-a)^2\big).
\end{aligned}$$

Two named factorizations appear constantly:

- **Sophie Germain's Identity:** $a^4 + 4b^4 = (a^2 + 2b^2 - 2ab)(a^2 + 2b^2 + 2ab)$.
- **Simon's Favorite Factoring Trick (SFFT):** $xy + ax + by + ab = (x+b)(y+a)$.

Example 1.1. Show that $n^4 + 4$ is composite for every integer $n \geq 2$.

Solution. By Sophie Germain with $a = n$, $b = 1$: $n^4 + 4 = (n^2 - 2n + 2)(n^2 + 2n + 2)$. For $n \geq 2$ both factors satisfy $n^2 - 2n + 2 = (n-1)^2 + 1 \geq 2$, so the number is composite. □

1.2 Sum and Product Notation, Telescoping, and Determinants

For a sequence $a_1, \ldots, a_n$ we write

$$\sum_{i=1}^{n} a_i = a_1 + a_2 + \cdots + a_n, \qquad \prod_{i=1}^{n} a_i = a_1 a_2 \cdots a_n.$$

Key manipulations include swapping the order of double sums $\sum_i \sum_j a_{ij} = \sum_j \sum_i a_{ij}$, index shifts, **telescoping**:

$$\sum_{k=1}^{n} \big(f(k+1) - f(k)\big) = f(n+1) - f(1), \qquad \prod_{k=1}^{n} \frac{f(k+1)}{f(k)} = \frac{f(n+1)}{f(1)},$$

and **Abel summation (summation by parts)**: with partial sums $A_k = \sum_{j=1}^{k} a_j$,

$$\sum_{k=1}^{n} a_k b_k = A_n b_n + \sum_{k=1}^{n-1} A_k (b_k - b_{k+1}).$$

Theorem 1.2 (Vandermonde Determinant)**.** *For any $x_1, \dots, x_n$,*

$$\det \begin{pmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & \cdots & x_2^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^{n-1} \end{pmatrix} = \prod_{1 \le i < j \le n} (x_j - x_i).$$

Proof. The determinant is a polynomial of total degree $\binom{n}{2}$ in $x_1, \dots, x_n$ that vanishes whenever $x_i = x_j$ (since two rows become identical). Hence it is divisible by $\prod_{i<j}(x_j - x_i)$. Comparing the coefficient of the main diagonal monomial $x_2^1 x_3^2 \cdots x_n^{n-1}$ on both sides shows the constant factor is 1. □

Example 1.3. Compute $\displaystyle\sum_{k=1}^{n} \frac{1}{k(k+1)}$.

Solution. Since $\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}$, the sum telescopes to $1 - \frac{1}{n+1} = \frac{n}{n+1}$. □

Practice Problem 1.4. Prove that $\displaystyle\prod_{k=2}^{n} \left(1 - \frac{1}{k^2}\right) = \frac{n+1}{2n}$.

Solution. Write $1 - \frac{1}{k^2} = \frac{(k-1)(k+1)}{k^2}$. Then

$$\prod_{k=2}^{n} \frac{(k-1)(k+1)}{k^2} = \frac{\prod_{k=2}^{n}(k-1) \cdot \prod_{k=2}^{n}(k+1)}{\left(\prod_{k=2}^{n} k\right)^2} = \frac{(n-1)! \cdot \frac{(n+1)!}{2}}{(n!)^2} = \frac{n+1}{2n}.$$ □

2 Functions: Injectivity, Surjectivity, and Functional Equations

Definition 2.1. A **function** $f\colon A \to B$ assigns to each $a \in A$ exactly one element $f(a) \in B$.

- f is **injective** (one-to-one) if $f(x) = f(y) \implies x = y$.
- f is **surjective** (onto) if for every $b \in B$ there is $a \in A$ with $f(a) = b$.
- f is **bijective** if it is both; then an inverse f^{-1} exists.
- $f\colon A \to A$ is an **involution** if $f(f(x)) = x$ for all $x \in A$. Every involution is automatically bijective.

2.1 Cauchy's Functional Equation and Classical Variants

Theorem 2.2 (Cauchy)**.** *Suppose $f\colon \mathbb{R} \to \mathbb{R}$ satisfies $f(x+y) = f(x) + f(y)$ for all x, y. Then $f(qx) = qf(x)$ for all $q \in \mathbb{Q}$; in particular $f(q) = cq$ on $\mathbb{Q}$ where $c = f(1)$. If in addition f is continuous, or monotonic, or bounded on some interval, or nonnegative for $x \ge 0$, then $f(x) = cx$ for all real x.*

Proof. Setting $x = y = 0$ gives $f(0) = 0$; setting $y = -x$ gives $f(-x) = -f(x)$. By induction $f(nx) = nf(x)$ for $n \in \mathbb{N}$, hence for all $n \in \mathbb{Z}$. For $q = m/n$, $nf(qx) = f(nqx) = f(mx) = mf(x)$, so $f(qx) = \frac{m}{n} f(x)$. Thus $f(q) = cq$ on $\mathbb{Q}$ with $c = f(1)$.

Now assume f is nondecreasing (if $f(x) \ge 0$ for $x \ge 0$, then $x \ge y \implies f(x) - f(y) = f(x-y) \ge 0$, reducing to this case). Fix $x \in \mathbb{R}$ and take rationals $p_n \uparrow x$, $q_n \downarrow x$. Then

$c\,p_n = f(p_n) \le f(x) \le f(q_n) = c\,q_n$. Letting $n \to \infty$ forces $f(x) = cx$. If instead f is bounded on an interval $[a, b]$, say $|f| \le M$ there, then $g(x) = f(x) - cx$ is additive, vanishes on $\mathbb{Q}$, and is bounded on $[a, b]$; additivity and $g(\mathbb{Q}) = 0$ imply g is bounded on all of $\mathbb{R}$ (translate by rationals). A nonzero value $g(x_0) \neq 0$ would give $g(nx_0) = ng(x_0)$ unbounded, contradiction. Hence $g \equiv 0$. □

Remark 2.3 (Classical Variants and Hamel Bases)**.** Under regularity hypotheses (continuity, monotonicity, or positivity):

(1) **Exponential Cauchy:** $f(x + y) = f(x)f(y) \implies f \equiv 0$ or $f(x) = a^x$ (take $\ln f(x)$ when $f > 0$).

(2) **Logarithmic Cauchy:** $f(xy) = f(x) + f(y)$ $(x, y > 0) \implies f(x) = c \ln x$ (set $g(t) = f(e^t)$).

(3) **Multiplicative Cauchy:** $f(xy) = f(x)f(y)$ $(x, y > 0) \implies f \equiv 0$ or $f(x) = x^c$.

(4) **Jensen's Equation:** $f\big(\frac{x+y}{2}\big) = \frac{f(x)+f(y)}{2} \implies g(x) := f(x) - f(0)$ is additive.

Without regularity, pathological solutions exist via Hamel bases of $\mathbb{R}$ over $\mathbb{Q}$. On olympiads, one derives regularity from the problem (e.g., $f(x^2) = f(x)^2 \ge 0$ gives $f \ge 0$ on positives, hence monotone).

2.2 The Olympiad Functional Equation Playbook

Standard moves in functional equations:

- **Substitute special values:** $x = y = 0$, $y = 0$, $y = -x$, or values that make arguments equal.
- **Prove injectivity or surjectivity:** If $f(\text{expr}) = x$, then f is surjective; if $f(x)$ is isolated on one side, fixing the other variables proves injectivity, allowing you to cancel outer f's ($f(A) = f(B) \implies A = B$).
- **Exploit symmetry and associativity:** Swap $x \leftrightarrow y$ or expand $f((x + 1)^2)$ in two ways.

Practice Problem 2.4. Find all $f\colon \mathbb{R} \to \mathbb{R}$ with $f(x^2 - y^2) = xf(x) - yf(y)$ for all x, y.

Solution. Put $x = y$: $f(0) = 0$. Put $y = 0$: $f(x^2) = xf(x)$; replacing $x \to -x$ gives $xf(x) = -xf(-x)$, so f is odd. The equation becomes $f(x^2-y^2) = f(x^2)-f(y^2)$, i.e. $f(a-b) = f(a)-f(b)$ for $a, b \ge 0$, which with oddness gives full additivity: $f(u + v) = f(u) + f(v)$ for all reals. Also $f(x^2) = xf(x)$; expand $f((x + 1)^2) = (x + 1)f(x + 1)$ using additivity:

$$f(x^2) + 2f(x) + f(1) = (x + 1)\big(f(x) + f(1)\big) = xf(x) + f(x) + xf(1) + f(1).$$

Since $f(x^2) = xf(x)$, this simplifies to $f(x) = xf(1)$. So $f(x) = cx$, and all such functions check. □

3 Polynomials

3.1 One Variable: Division, Vieta, and Arithmetic Properties

A polynomial of degree n is $P(x) = a_n x^n + \cdots + a_0$ with $a_n \neq 0$.

Theorem 3.1 (Division Algorithm)**.** *For polynomials P, D with $D \neq 0$ over a field, there exist unique Q, R with $P = QD + R$ and $\deg R < \deg D$ (or $R = 0$).*

Corollary 3.2 (Factor Theorem)**.** $P(r) = 0 \iff (x - r) \mid P(x)$. *Consequently a nonzero polynomial of degree n has at most n roots; two degree-$\leq n$ polynomials agreeing at $n+1$ points are identical.*

Proof. Divide P by $x-r$: $P(x) = (x-r)Q(x)+P(r)$. The rest follows by induction on the number of roots. □

Theorem 3.3 (Fundamental Theorem of Algebra)**.** *Every nonconstant polynomial with complex coefficients has a complex root; hence P of degree n factors as $a_n \prod_{i=1}^{n}(x - r_i)$ over $\mathbb{C}$.*

Proof sketch. $|P(z)| \to \infty$ as $|z| \to \infty$, so $|P|$ attains a minimum at some z_0. If $P(z_0) \neq 0$, write $P(z_0 + w) = P(z_0) + cw^k + O(w^{k+1})$ with $c \neq 0$ and choose the direction of w so that cw^k points opposite to $P(z_0)$; for small $|w|$ this strictly decreases $|P|$, a contradiction (d'Alembert's argument). □

Theorem 3.4 (Vieta's Relations)**.** *If $P(x) = a_n \prod_{i=1}^{n}(x - r_i) = \sum_{k=0}^{n} a_k x^k$, then for each $1 \leq k \leq n$,*

$$e_k(r_1, \ldots, r_n) := \sum_{i_1 < \cdots < i_k} r_{i_1} \cdots r_{i_k} = (-1)^k \frac{a_{n-k}}{a_n}.$$

Proof. Expand the product $\prod(x - r_i)$: the coefficient of x^{n-k} collects one factor $-r_i$ from exactly k of the brackets, giving $(-1)^k e_k$. Compare coefficients. □

Proposition 3.5. *If $P \in \mathbb{Z}[x]$ and $a, b \in \mathbb{Z}$, then $(a - b) \mid P(a) - P(b)$.*

Proof. $P(a) - P(b) = \sum_k c_k(a^k - b^k)$ and $(a - b) \mid a^k - b^k$ for each k. □

Theorem 3.6 (Rational Root Theorem)**.** *If p/q (in lowest terms) is a root of $a_n x^n + \cdots + a_0 \in \mathbb{Z}[x]$, then $p \mid a_0$ and $q \mid a_n$.*

Example 3.7. Let r, s, t be the roots of $x^3 - 2x^2 + 3x - 4$. Compute $r^2 + s^2 + t^2$.

Solution. By Vieta, $r+s+t = 2$, $rs+st+tr = 3$. Then $r^2+s^2+t^2 = (r+s+t)^2-2(rs+st+tr) = 4 - 6 = -2$. □

Practice Problem 3.8. Let $P \in \mathbb{Z}[x]$ and suppose $P(a) = P(b) = P(c) = 2$ for distinct integers a, b, c. Prove $P(x) = 3$ has no integer solution.

Solution. Write $P(x) - 2 = (x - a)(x - b)(x - c)Q(x)$ with $Q \in \mathbb{Z}[x]$ (divide successively; integer coefficients persist since leading divisions are by monic factors). If $P(d) = 3$ for $d \in \mathbb{Z}$, then $(d - a)(d - b)(d - c)Q(d) = 1$, so each of $d - a$, $d - b$, $d - c \in \{1, -1\}$. By pigeonhole two of them are equal, contradicting a, b, c distinct. □

3.2 Lagrange Interpolation and Irreducibility Criteria

Theorem 3.9 (Lagrange Interpolation)**.** *Given $n + 1$ distinct points $x_0, x_1, \ldots, x_n$ and values $y_0, y_1, \ldots, y_n$ over a field, there exists a unique polynomial $P(x)$ of degree $\leq n$ such that $P(x_i) = y_i$ for all i:*

$$P(x) = \sum_{i=0}^{n} y_i \prod_{j \neq i} \frac{x - x_j}{x_i - x_j}.$$

In particular, the coefficient of x^n in $P(x)$ is $\sum_{i=0}^{n} \frac{y_i}{\prod_{j \neq i}(x_i - x_j)}$.

Proof. Each basis polynomial $L_i(x) = \prod_{j \neq i} \frac{x - x_j}{x_i - x_j}$ has degree n and satisfies $L_i(x_k) = \delta_{ik}$. Thus $P(x) = \sum_{i=0}^{n} y_i L_i(x)$ has degree $\leq n$ and satisfies $P(x_k) = y_k$. Uniqueness follows from the Factor Theorem applied to the difference of two such polynomials. □

Theorem 3.10 (Gauss's Lemma, Eisenstein, Perron, and Cohn)**.** *Let $P(x) = a_n x^n + \cdots + a_0 \in \mathbb{Z}[x]$.*

(a) ***(Gauss's Lemma)*** *P is irreducible over $\mathbb{Q}$ iff it cannot be factored into two nonconstant polynomials in $\mathbb{Z}[x]$.*

(b) ***(Eisenstein's Criterion)*** *If there exists a prime p such that $p \nmid a_n$, $p \mid a_i$ for all $0 \leq i < n$, and $p^2 \nmid a_0$, then $P(x)$ is irreducible over $\mathbb{Q}$.*

(c) ***(Perron's Criterion)*** *If $a_n = 1$, $a_0 \neq 0$, and $|a_{n-1}| > 1 + |a_{n-2}| + \cdots + |a_0|$, then $P(x)$ is irreducible over $\mathbb{Q}$.*

(d) ***(Cohn's Criterion)*** *If a prime p is written in base $b \geq 2$ as $p = \sum_{k=0}^{n} d_k b^k$ $(0 \leq d_k < b)$, then $P(x) = \sum_{k=0}^{n} d_k x^k$ is irreducible over $\mathbb{Q}$.*

Proof of (b). Suppose $P(x) = G(x)H(x)$ in $\mathbb{Z}[x]$ with $\deg G, \deg H \geq 1$. Modulo p, $\overline{P}(x) = \overline{a_n} x^n = \overline{G}(x)\overline{H}(x)$ in $\mathbb{F}_p[x]$. By unique factorization in $\mathbb{F}_p[x]$, both $\overline{G}(x)$ and $\overline{H}(x)$ are monomials in x, so their constant terms g_0, h_0 are divisible by p. Then $p^2 \mid g_0 h_0 = a_0$, contradicting $p^2 \nmid a_0$. □

Practice Problem 3.11. Let $P(x)$ be a polynomial of degree n such that $P(k) = \frac{k}{k+1}$ for $k = 0, 1, \ldots, n$. Find $P(n+1)$.

Solution. Consider $Q(x) = (x+1)P(x) - x$. Then $\deg Q = n+1$ and $Q(k) = 0$ for $k = 0, 1, \ldots, n$. Thus

$$(x+1)P(x) - x = c\, x(x-1)(x-2)\cdots(x-n).$$

Plugging in $x = -1$ gives $1 = c(-1)^{n+1}(n+1)!$, so $c = \frac{(-1)^{n+1}}{(n+1)!}$. Evaluating at $x = n+1$ yields

$$(n+2)P(n+1) - (n+1) = (-1)^{n+1} \implies P(n+1) = \frac{n + 1 + (-1)^{n+1}}{n+2}.$$ □

3.3 Symmetric Polynomials and Chebyshev Polynomials

A polynomial $P(x_1, \ldots, x_n)$ is **symmetric** if it is invariant under every permutation of the variables, and **homogeneous of degree** d if every monomial has total degree d.

Theorem 3.12 (Fundamental Theorem of Symmetric Polynomials)**.** *Every symmetric polynomial in $x_1, \ldots, x_n$ can be written uniquely as a polynomial in the elementary symmetric polynomials $e_1, \ldots, e_n$.*

Proof sketch. Order monomials lexicographically. Given a symmetric P with leading monomial $x_1^{a_1} \cdots x_n^{a_n}$ (necessarily $a_1 \geq \cdots \geq a_n$), subtract $c\, e_1^{a_1 - a_2} e_2^{a_2 - a_3} \cdots e_n^{a_n}$, which has the same leading monomial. The leading term strictly decreases; iterate until zero. □

Newton's Identities. With power sums $p_k = \sum x_i^k$ and $e_j = 0$ for $j > n$:

$$p_k - e_1 p_{k-1} + e_2 p_{k-2} - \cdots + (-1)^{k-1} e_{k-1} p_1 + (-1)^k k\, e_k = 0 \quad (k \geq 1).$$

Definition 3.13 (Chebyshev Polynomials)**.** The **Chebyshev polynomials** $T_n(x)$ are defined by $T_0(x) = 1$, $T_1(x) = x$, and $T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x)$. By induction and $\cos(n+1)\theta + \cos(n-1)\theta = 2\cos\theta\cos n\theta$, they satisfy

$$T_n(\cos\theta) = \cos(n\theta), \qquad T_n\left(\frac{z+z^{-1}}{2}\right) = \frac{z^n + z^{-n}}{2}, \qquad T_m(T_n(x)) = T_{mn}(x).$$

Practice Problem 3.14. Factor $a^3+b^3+c^3-3abc$, deduce that $a+b+c \mid a^3+b^3+c^3-3abc$ for all integers a, b, c, and characterize when $a^3+b^3+c^3 = 3abc$ for real a, b, c.

Solution. Viewing $P(a) = a^3+b^3+c^3-3abc$ as a polynomial in a, substitute $a = -(b+c)$: we get $-(b+c)^3 + b^3 + c^3 + 3bc(b+c) = 0$. Hence $(a+b+c) \mid P$, and polynomial division yields

$$a^3+b^3+c^3-3abc = (a+b+c)(a^2+b^2+c^2-ab-bc-ca) = \frac{1}{2}(a+b+c)\big((a-b)^2+(b-c)^2+(c-a)^2\big).$$

This gives the divisibility over $\mathbb{Z}$, and shows $a^3+b^3+c^3 = 3abc \iff a+b+c = 0$ or $a = b = c$ over $\mathbb{R}$. □

4 Complex Numbers and Trigonometry

Definition 4.1. $\mathbb{C} = \{a+bi : a, b \in \mathbb{R}\}$, $i^2 = -1$. For $z = a+bi$: the conjugate is $\bar{z} = a - bi$, the modulus $|z| = \sqrt{a^2+b^2} = \sqrt{z\bar{z}}$, and the polar form is $z = r(\cos\theta + i\sin\theta) = re^{i\theta}$.

Theorem 4.2 (Euler / de Moivre and Triangle Identities)**.**

(a) $e^{i\theta} = \cos\theta + i\sin\theta$, *hence* $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$.

(b) If $A+B+C = \pi$, *then:*

$$\tan A + \tan B + \tan C = \tan A\tan B\tan C, \qquad \sin A + \sin B + \sin C = 4\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2},$$
$$\cot\frac{A}{2} + \cot\frac{B}{2} + \cot\frac{C}{2} = \cot\frac{A}{2}\cot\frac{B}{2}\cot\frac{C}{2}, \quad \cos A + \cos B + \cos C = 1 + 4\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}.$$

Proof. De Moivre follows by induction from $(\cos\alpha+i\sin\alpha)(\cos\beta+i\sin\beta) = \cos(\alpha+\beta)+i\sin(\alpha+\beta)$. For tan: $\tan(A+B) = -\tan C \implies \frac{\tan A+\tan B}{1-\tan A\tan B} = -\tan C$, which rearranges to the identity. □

Roots of unity. The solutions of $z^n = 1$ are $\omega^k = e^{2\pi ik/n}$, $k = 0, \ldots, n-1$. They satisfy

$$z^n - 1 = \prod_{k=0}^{n-1}(z - \omega^k), \qquad \sum_{k=0}^{n-1}\omega^{km} = \begin{cases} n & n \mid m, \\ 0 & n \nmid m, \end{cases}$$

the latter being the **roots of unity filter**, which extracts every n-th coefficient of $P(x) = \sum a_m x^m$:

$$\sum_{m \equiv r \pmod n} a_m = \frac{1}{n}\sum_{k=0}^{n-1}\omega^{-kr}P(\omega^k).$$

Example 4.3. Evaluate $\cos\frac{2\pi}{5} + \cos\frac{4\pi}{5}$.

Solution. The five 5th roots of unity sum to 0. Taking real parts: $1 + 2\cos\frac{2\pi}{5} + 2\cos\frac{4\pi}{5} = 0$, so the sum equals $-\frac{1}{2}$. □

Practice Problem 4.4. Prove that $\displaystyle\sum_{k\equiv 0 \pmod 3} \binom{n}{k} = \frac{2^n + 2\cos\frac{n\pi}{3}}{3}$ and $\displaystyle\prod_{k=1}^{n-1} \sin\frac{k\pi}{n} = \frac{n}{2^{n-1}}$.

Solution. Let $\omega = e^{2\pi i/3}$. By the roots of unity filter applied to $(1+x)^n = \sum_k \binom{n}{k} x^k$,

$$\sum_{3\mid k} \binom{n}{k} = \frac{(1+1)^n + (1+\omega)^n + (1+\omega^2)^n}{3}.$$

Now $1+\omega = e^{i\pi/3}$ and $1+\omega^2 = e^{-i\pi/3}$, so $(1+\omega)^n + (1+\omega^2)^n = 2\cos\frac{n\pi}{3}$, giving the first claim. For the product, divide $z^n - 1 = \prod_{k=0}^{n-1}(z - e^{2\pi i k/n})$ by $z-1$, set $z=1$, and take moduli using $|1 - e^{2i\theta}| = 2|\sin\theta|$. □

5 Inequalities

5.1 AM–GM and Cauchy–Schwarz

Theorem 5.1 (AM–GM)**.** *For nonnegative reals $a_1, \ldots, a_n$,*

$$\frac{a_1 + \cdots + a_n}{n} \geq \sqrt[n]{a_1 \cdots a_n},$$

with equality iff all a_i are equal.

Proof (Cauchy's forward–backward induction). Case $n = 2$: $(\sqrt{a_1} - \sqrt{a_2})^2 \geq 0$. *Powers of 2:* if the claim holds for n, apply it to two blocks of n and then the case $n = 2$ to their averages; this proves it for $2n$, hence all $n = 2^k$. *Backward step:* if the claim holds for n, apply it for n to $a_1, \ldots, a_{n-1}, A$ where $A = \frac{a_1 + \cdots + a_{n-1}}{n-1}$. The arithmetic mean of these n numbers is again A, so $A \geq (a_1 \cdots a_{n-1} A)^{1/n}$, i.e. $A^{n-1} \geq a_1 \cdots a_{n-1}$, which is the claim for $n-1$. Combining, AM–GM holds for all n. Equality tracing shows all terms equal. □

Theorem 5.2 (Cauchy–Schwarz)**.** *For reals a_i, b_i,*

$$\left(\sum_{i=1}^n a_i b_i\right)^2 \leq \left(\sum_{i=1}^n a_i^2\right)\left(\sum_{i=1}^n b_i^2\right),$$

with equality iff the vectors (a_i) and (b_i) are proportional.

Proof. The quadratic $q(t) = \sum_i (a_i t + b_i)^2 = t^2 \sum a_i^2 + 2t \sum a_i b_i + \sum b_i^2$ is ≥ 0 for all t, so its discriminant is ≤ 0, which is the inequality. Equality means q has a real root t_0, i.e. $b_i = -t_0 a_i$ for all i. □

Corollary 5.3 (Cauchy–Schwarz in Engel form / Titu's Lemma)**.** *For reals x_i and positive y_i:* $\displaystyle\sum \frac{x_i^2}{y_i} \geq \frac{(\sum x_i)^2}{\sum y_i}$.

Proof. Apply Cauchy–Schwarz to $a_i = x_i/\sqrt{y_i}$, $b_i = \sqrt{y_i}$. □

Example 5.4 (Nesbitt's Inequality)**.** For $a, b, c > 0$: $\displaystyle\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2}$.

Solution. By Engel form,

$$\sum \frac{a}{b+c} = \sum \frac{a^2}{a(b+c)} \geq \frac{(a+b+c)^2}{2(ab+bc+ca)} \geq \frac{3}{2},$$

the last step because $(a+b+c)^2 \geq 3(ab+bc+ca)$, i.e. $a^2+b^2+c^2 \geq ab+bc+ca$, which follows from summing $\frac{1}{2}\big((a-b)^2+(b-c)^2+(c-a)^2\big) \geq 0$. □

5.2 Rearrangement, Chebyshev, and Schur's Inequalities

Theorem 5.5 (Rearrangement and Chebyshev Inequalities)**.** *Let $a_1 \leq a_2 \leq \cdots \leq a_n$ and $b_1 \leq b_2 \leq \cdots \leq b_n$ be real numbers. For any permutation σ of $\{1,\ldots,n\}$,*

$$\sum_{i=1}^{n} a_i b_{n+1-i} \leq \sum_{i=1}^{n} a_i b_{\sigma(i)} \leq \sum_{i=1}^{n} a_i b_i, \qquad \text{and} \qquad \frac{1}{n}\sum_{i=1}^{n} a_i b_i \geq \left(\frac{1}{n}\sum_{i=1}^{n} a_i\right)\left(\frac{1}{n}\sum_{i=1}^{n} b_i\right).$$

Proof. If σ has an inversion $i<j$ with $\sigma(i)>\sigma(j)$, swapping $\sigma(i)$ and $\sigma(j)$ changes the sum by $(a_j-a_i)(b_{\sigma(i)}-b_{\sigma(j)}) \geq 0$. Uncrossing all inversions reaches $\sigma = \mathrm{id}$. Averaging the n cyclic shifts gives Chebyshev's inequality. □

Theorem 5.6 (Schur's Inequality)**.** *For nonnegative reals x,y,z and $r>0$,*

$$x^r(x-y)(x-z)+y^r(y-z)(y-x)+z^r(z-x)(z-y) \geq 0,$$

with equality iff $x=y=z$ or two variables are equal and the third is 0. For $r=1$: $x^3+y^3+z^3+3xyz \geq \sum_{sym} x^2y$, or $p^3-4pq+9r \geq 0$ ($p=\sum x, q=\sum xy, r=xyz$).

Proof. WLOG assume $x \geq y \geq z \geq 0$. Rewrite the left-hand side as

$$(x-y)\big(x^r(x-z)-y^r(y-z)\big)+z^r(z-x)(z-y).$$

Since $x^r \geq y^r \geq 0$ and $x-z \geq y-z \geq 0$, the first term is ≥ 0; since $z-x \leq 0$ and $z-y \leq 0$, the second term is also ≥ 0. □

Remark 5.7 (Homogenization and Ravi Substitution)**.** Given a constraint like $a+b+c=1$, multiply terms by powers of $a+b+c$ to make the inequality homogeneous. When a,b,c are the sides of a triangle, the **Ravi substitution** $a=y+z, b=z+x, c=x+y$ $(x,y,z>0)$ eliminates the triangle inequality constraints.

Practice Problem 5.8 (IMO 2000/2)**.** Let $a,b,c>0$ with $abc=1$. Prove that

$$\left(a-1+\frac{1}{b}\right)\left(b-1+\frac{1}{c}\right)\left(c-1+\frac{1}{a}\right) \leq 1$$

Solution. Substitute $a=x/y, b=y/z, c=z/x$ for $x,y,z>0$. Clearing denominators, the inequality becomes $(x+y-z)(y+z-x)(z+x-y) \leq xyz$. At most one factor on the left is ≤ 0 (as the sum of any two is positive); if one is ≤ 0, the product is $\leq 0 < xyz$. Otherwise expanding gives $x^3+y^3+z^3+3xyz \geq \sum_{\mathrm{sym}} x^2y$, which is Schur for $r=1$. □

5.3 Jensen's, Hölder's, and Minkowski's Inequalities [Advanced]

Definition 5.9. f is **convex** on an interval I if $f(\lambda x + (1-\lambda)y) \le \lambda f(x) + (1-\lambda)f(y)$ for all $x, y \in I$, $\lambda \in [0,1]$; **concave** if the inequality reverses. If $f'' \ge 0$ on I then f is convex.

Theorem 5.10 (Jensen)**.** *If f is convex on I, $x_i \in I$, and $w_i > 0$ with $\sum w_i = 1$, then*

$$f\Big(\sum_{i=1}^{n} w_i x_i\Big) \le \sum_{i=1}^{n} w_i f(x_i).$$

Proof. Induct on n; the case $n = 2$ is the definition. For $n > 2$ write $\sum_{i=1}^{n} w_i x_i = w_n x_n + (1-w_n)y$ where $y = \sum_{i<n} \frac{w_i}{1-w_n} x_i \in I$. Then by convexity and the induction hypothesis,

$$f\Big(\sum w_i x_i\Big) \le w_n f(x_n) + (1-w_n) f(y) \le w_n f(x_n) + (1-w_n) \sum_{i<n} \tfrac{w_i}{1-w_n} f(x_i) = \sum w_i f(x_i). \quad \square$$

Theorem 5.11 (Hölder and Minkowski)**.** *For positive reals and exponents $p, q > 1$ with $\frac{1}{p} + \frac{1}{q} = 1$:*

$$\sum_{i=1}^{n} a_i b_i \le \Big(\sum a_i^p\Big)^{1/p} \Big(\sum b_i^q\Big)^{1/q}, \qquad \Big(\sum (a_i + b_i)^p\Big)^{1/p} \le \Big(\sum a_i^p\Big)^{1/p} + \Big(\sum b_i^p\Big)^{1/p}.$$

In three-sequence olympiad form: $\big(\sum a_i\big)\big(\sum b_i\big)\big(\sum c_i\big) \ge \big(\sum \sqrt[3]{a_i b_i c_i}\big)^3$.

Proof. For Hölder, normalize $\sum a_i^p = \sum b_i^q = 1$. By weighted AM–GM, $a_i b_i = (a_i^p)^{1/p}(b_i^q)^{1/q} \le \frac{a_i^p}{p} + \frac{b_i^q}{q}$; summing gives $\sum a_i b_i \le \frac{1}{p} + \frac{1}{q} = 1$. For Minkowski, let $s_i = a_i + b_i$; by Hölder, $\sum s_i^p = \sum a_i s_i^{p-1} + \sum b_i s_i^{p-1} \le \big[(\sum a_i^p)^{1/p} + (\sum b_i^p)^{1/p}\big](\sum s_i^p)^{1/q}$, and divide by $(\sum s_i^p)^{1/q}$. $\square$

5.4 Muirhead, Karamata, SOS, and the *uvw* Method [Advanced]

Theorem 5.12 (Muirhead, Karamata, SOS, and Teichmüller's *uvw* Theorem)**.**

(a) ***(Muirhead & Karamata)*** *Say $\alpha \succ \beta$ if $\alpha_1 \ge \cdots \ge \alpha_n$, $\beta_1 \ge \cdots \ge \beta_n$, $\sum_{i=1}^{k} \alpha_i \ge \sum_{i=1}^{k} \beta_i$ $(k < n)$, and $\sum_{i=1}^{n} \alpha_i = \sum_{i=1}^{n} \beta_i$. If $\alpha \succ \beta$, then $\sum_{\sigma \in S_n} x_{\sigma(1)}^{\alpha_1} \cdots x_{\sigma(n)}^{\alpha_n} \ge \sum_{\sigma \in S_n} x_{\sigma(1)}^{\beta_1} \cdots x_{\sigma(n)}^{\beta_n}$ for $x_i > 0$, and $\sum_{i=1}^{n} f(\alpha_i) \ge \sum_{i=1}^{n} f(\beta_i)$ for any convex f (proved via Abel summation on the nonincreasing secant slopes $c_i = \frac{f(\alpha_i) - f(\beta_i)}{\alpha_i - \beta_i}$).*

(b) ***(SOS Method)*** *$\sum_{cyc} S_a (b-c)^2 \ge 0$ holds if $S_a, S_b, S_c \ge 0$, or if $S_a, S_c \ge 0$ and $S_a + 2S_b \ge 0, S_c + 2S_b \ge 0$.*

(c) ***(uvw / pqr Theorem)*** *For $a, b, c \ge 0$ with fixed $p = a + b + c$ and $q = ab + bc + ca$, the product $r = abc$ attains its extrema only when* two variables are equal *or* one variable is 0 *(since a, b, c are the real roots of $t^3 - pt^2 + qt = r$, and shifting the horizontal line $y = r$ reaches an extremum when the line is tangent to the cubic—a double root—or hits $r = 0$). Thus any inequality $f(p, q) + Cr \ge 0$ linear in r need only be checked for $a = b$ and $c = 0$.*

Practice Problem 5.13. Let $a, b, c \ge 0$ with $a + b + c = 3$. (a) If $a, b, c > 0$, prove $\dfrac{1}{a^2} + \dfrac{1}{b^2} + \dfrac{1}{c^2} \ge a^2 + b^2 + c^2$. (b) Prove $a^2 + b^2 + c^2 + abc \ge 4$.

Solution. (a) Use the *tangent line trick* on $f(x) = \frac{1}{x^2} - x^2$ at $x = 1$: $f(1) = 0$, $f'(1) = -4$, and

$$\frac{1}{x^2} - x^2 - \big(-4(x-1)\big) = \frac{(x-1)^2(-x^2+2x+1)}{x^2} \geq 0 \quad \text{for } x \in (0, 1+\sqrt{2}].$$

If $a, b, c \leq 1+\sqrt{2}$, summing gives $\sum f(a) \geq -4(a+b+c-3) = 0$. If say $c > 1+\sqrt{2}$, then $a+b < 2-\sqrt{2}$, so $\min(a,b) < 1 - \frac{\sqrt{2}}{2} < 0.3$ and $\frac{1}{a^2} + \frac{1}{b^2} > \frac{1}{0.09} > 11 > 9 \geq a^2+b^2+c^2$.

(b) In p, q, r form with $p = 3$, $a^2+b^2+c^2+abc-4 = 5-2q+r \geq 0$, which is linear in r. By *uvw*: if $c = 0$, $a^2+b^2 \geq \frac{(a+b)^2}{2} = \frac{9}{2} > 4$; if $a = b = x$ and $c = 3-2x$ ($x \in [0, 3/2]$), then $2x^2 + (3-2x)^2 + x^2(3-2x) - 4 = (x-1)^2(5-2x) \geq 0$. □

6 Floor Functions, Beatty's Theorem, and Sequence Asymptotics

Theorem 6.1 (Hermite's Identity, Beatty's Theorem, and Telescoping Asymptotics)**.**

(a) ***(Hermite)*** *For $x \in \mathbb{R}$ and $n \in \mathbb{N}$, $\sum_{k=0}^{n-1} \lfloor x + \frac{k}{n} \rfloor = \lfloor nx \rfloor$.*

(b) ***(Beatty)*** *If $\alpha, \beta > 1$ are irrational with $\frac{1}{\alpha} + \frac{1}{\beta} = 1$, then $A = \{\lfloor n\alpha \rfloor : n \in \mathbb{N}\}$ and $B = \{\lfloor m\beta \rfloor : m \in \mathbb{N}\}$ partition $\mathbb{N}$.*

(c) ***(Telescoping Powers)*** *For $x_{n+1} = x_n + cx_n^{-\alpha}$ ($c > 0$), raising to power $\alpha+1$ makes $x_{n+1}^{\alpha+1} - x_n^{\alpha+1} \to (\alpha+1)c$, yielding $x_n \sim \big((\alpha+1)cn\big)^{1/(\alpha+1)}$.*

Proof of (a) and (b). (a) Write $x = m + r/n + \delta$ with $m = \lfloor x \rfloor$, $r \in \{0, \ldots, n-1\}$, $0 \leq \delta < 1/n$; both sides equal $nm + r$. (b) For any integer $k \geq 1$, $\{k/\alpha\} + \{k/\beta\} = 1$ since $k/\alpha + k/\beta = k \in \mathbb{Z}$ and both are irrational. Thus $\lfloor k/\alpha \rfloor + \lfloor k/\beta \rfloor = k-1$. Subtracting this for k and $k+1$ gives $\big(\lfloor \frac{k+1}{\alpha} \rfloor - \lfloor \frac{k}{\alpha} \rfloor\big) + \big(\lfloor \frac{k+1}{\beta} \rfloor - \lfloor \frac{k}{\beta} \rfloor\big) = 1$, so k lies in exactly one of A, B. □

Practice Problem 6.2. (a) Let $x_1 = 1$ and $x_{n+1} = x_n + \frac{1}{x_n}$. Find $\lim_{n\to\infty} \frac{x_n}{\sqrt{n}}$. (b) Prove that $\lfloor (2+\sqrt{3})^n \rfloor$ is odd for every positive integer n.

Solution. (a) Squaring gives $x_{n+1}^2 - x_n^2 = 2 + x_n^{-2} > 2$, so summing from 1 to $n-1$ gives $x_n^2 \geq 2n-1$. Substituting this back gives $x_n^2 = 2n - 1 + \sum_{k=1}^{n-1} x_k^{-2} = 2n + O(\ln n)$, so $x_n/\sqrt{n} \to \sqrt{2}$. (b) Let $u_n = (2+\sqrt{3})^n + (2-\sqrt{3})^n$. Then $u_0 = 2, u_1 = 4$, and $u_{n+1} = 4u_n - u_{n-1}$, so u_n is always an even integer. Since $0 < (2-\sqrt{3})^n < 1$, we have $\lfloor (2+\sqrt{3})^n \rfloor = u_n - 1$, which is odd. □

7 Generating Functions [Advanced]

Definition 7.1. The **ordinary generating function** of a sequence $(a_n)_{n\geq 0}$ is the formal power series $A(x) = \sum_{n\geq 0} a_n x^n$.

Key dictionary: addition of series $\leftrightarrow$ termwise addition; multiplication $\leftrightarrow$ convolution $\big(\sum a_n x^n\big)\big(\sum b_n x^n\big) = \sum_n \big(\sum_k a_k b_{n-k}\big) x^n$; $\frac{1}{1-x} = \sum x^n$; and the generalized binomial theorem $\frac{1}{(1-x)^k} = \sum_{n\geq 0} \binom{n+k-1}{k-1} x^n$.

Example 7.2 (Fibonacci Closed Form)**.** Let $F_0 = 0$, $F_1 = 1$, $F_{n+2} = F_{n+1} + F_n$. Then with $\varphi = \frac{1+\sqrt{5}}{2}$, $\psi = \frac{1-\sqrt{5}}{2}$: $F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}$.

Solution. Let $F(x) = \sum F_n x^n$. The recurrence gives $F(x) - x = xF(x) + x^2F(x)$, so $F(x) = \frac{x}{1-x-x^2} = \frac{1}{\sqrt{5}}\left(\frac{1}{1-\varphi x} - \frac{1}{1-\psi x}\right)$, and expanding each geometric series yields the formula. □

Practice Problem 7.3 (Sicherman Dice)**.** Show that one cannot relabel two six-sided dice with positive integers so that every pairwise sum $2, \ldots, 12$ is equally likely, but find the unique nonstandard positive-integer labeling (Sicherman dice) giving the same sum distribution as two standard dice.

Solution. A die with faces $a_1, \ldots, a_6$ has generating function $D(x) = \sum x^{a_i}$. A uniform distribution on the 11 sums $\{2, \ldots, 12\}$ is impossible since $11 \nmid 36 = D_1(1)D_2(1)$. For standard dice:

$$\left(x + \cdots + x^6\right)^2 = x^2(1+x)^2(1+x+x^2)^2(1-x+x^2)^2.$$

We must split these irreducible factors into $D_1(x)D_2(x)$ with $D_i(1) = 6$ and $D_i(0) = 0$. Since $(1+x)(1+x+x^2)\big|_{x=1} = 6$ and $(1-x+x^2)\big|_{x=1} = 1$, each die must receive $x(1+x)(1+x+x^2)$. Giving both factors of $(1-x+x^2)$ to D_2 yields

$$D_1(x) = x(1+x)(1+x+x^2) = x + 2x^2 + 2x^3 + x^4$$

$$D_2(x) = x(1+x)(1+x+x^2)(1-x+x^2)^2 = x + x^3 + x^4 + x^5 + x^6 + x^8$$

corresponding to faces $\{1, 2, 2, 3, 3, 4\}$ and $\{1, 3, 4, 5, 6, 8\}$. □

8 Calculus in Olympiads [Advanced]

Theorem 8.1 (Key Facts)**.**

(a) If $f'(x) \geq 0$ on an interval, f is nondecreasing there.

(b) ***(Rolle)*** *If f is differentiable and $f(a) = f(b)$, some $c \in (a, b)$ has $f'(c) = 0$. Consequently, between two real roots of a polynomial lies a root of its derivative; a polynomial with all real roots has all derivatives with all real roots.*

(c) If $f'' \geq 0$, then f is convex and lies above each tangent line: $f(x) \geq f(a) + f'(a)(x-a)$ (the tangent line trick*).*

(d) ***(Smoothing)*** *To extremize a symmetric expression under a constraint, examine what happens when two variables move toward (or away from) each other with their sum or product fixed.*

Proof of (c). Let $g(x) = f(x) - f(a) - f'(a)(x-a)$. Then $g(a) = 0$, $g'(a) = 0$, and $g'' = f'' \geq 0$ means g' is nondecreasing; so $g' \leq 0$ before a and $g' \geq 0$ after, hence g attains its minimum 0 at a. □

Practice Problem 8.2. (a) Let $x, y, z > 0$ with $x + y + z = 1$. Find the minimum of $\sum \frac{x}{1-x}$. (b) If $P(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0$ $(n \geq 2)$ has n real roots, prove $(n-1)a_{n-1}^2 \geq 2na_{n-2}$.

Solution. (a) $f(t) = \frac{t}{1-t}$ on $(0, 1)$ has $f''(t) = \frac{2}{(1-t)^3} > 0$. By Jensen, $\sum f(x_i) \geq 3f(1/3) = \frac{3}{2}$. (b) By Rolle's theorem applied $n-2$ times, $\frac{P^{(n-2)}(x)}{(n-2)!} = \frac{n(n-1)}{2}x^2 + (n-1)a_{n-1}x + a_{n-2}$ has two real roots, so its discriminant $(n-1)^2a_{n-1}^2 - 2n(n-1)a_{n-2} \geq 0$. □

Part II
Combinatorics

9 Basic Counting and Binomial Identities

Fundamental principles. If a procedure has m outcomes and an independent second stage has n, together there are mn (product rule); disjoint cases add (sum rule).

Definition 9.1. $n! = n(n-1)\cdots 1$ counts orderings of n objects; $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ counts k-element subsets of an n-element set; 2^n counts all subsets.

Theorem 9.2 (Binomial Theorem)**.** $(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}$.

Proof. Expanding the product of n brackets, each term chooses x from k brackets and y from the rest; there are $\binom{n}{k}$ ways to choose which. □

Core identities (each with a combinatorial proof worth internalizing):

$$\binom{n}{k} = \binom{n}{n-k}, \qquad \binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1} \quad \text{(Pascal)},$$

$$k\binom{n}{k} = n\binom{n-1}{k-1} \quad \text{(Leader)}, \qquad \sum_{k=r}^{n} \binom{k}{r} = \binom{n+1}{r+1} \quad \text{(Hockey-Stick)},$$

$$\sum_{k=0}^{r} \binom{m}{k}\binom{n}{r-k} = \binom{m+n}{r} \quad \text{(Vandermonde)}, \quad \#\{x_1 + \cdots + x_k = n,\ x_i \geq 0\} = \binom{n+k-1}{k-1} \quad \text{(Stars \& Bars)}$$

Example 9.3. How many lattice paths go from $(0,0)$ to (m,n) using unit steps right and up?

Solution. A path is a sequence of $m+n$ steps of which m are "right": $\binom{m+n}{m}$ paths. □

Practice Problem 9.4. Prove $\displaystyle\sum_{k=0}^{n} \binom{n}{k}^2 = \binom{2n}{n}$.

Solution. Count committees of n people chosen from n men and n women. Choosing k men and $n-k$ women can be done in $\binom{n}{k}\binom{n}{n-k} = \binom{n}{k}^2$ ways; summing over k counts all $\binom{2n}{n}$ committees. □

10 Mathematical Induction

Theorem 10.1 (Principle of Induction)**.** *Let $S \subseteq \mathbb{N}$ contain 1, and suppose $n \in S \implies n+1 \in S$. Then $S = \mathbb{N}$. Equivalent forms:* strong induction *(assume all of $1, \ldots, n$ to prove $n+1$) and* well-ordering *(every nonempty subset of $\mathbb{N}$ has a least element).*

Example 10.2. Prove that $\sum_{k=1}^{n} k^3 = \left(\frac{n(n+1)}{2}\right)^2$.

Solution. Base $n = 1$: both sides equal 1. Step: assuming the identity for n,

$$\sum_{k=1}^{n+1} k^3 = \frac{n^2(n+1)^2}{4} + (n+1)^3 = \frac{(n+1)^2\left(n^2+4n+4\right)}{4} = \left(\frac{(n+1)(n+2)}{2}\right)^2. \qquad \square$$

Practice Problem 10.3 (Strong Induction)**.** Every integer $n \geq 2$ is a product of primes.

Solution. Strong induction. If n is prime, done. Otherwise $n = ab$ with $2 \leq a, b < n$; by the strong induction hypothesis each of a, b is a product of primes, hence so is n. □

Remark 10.4. Powerful variants: strengthen the inductive hypothesis, induct on an unusual quantity (sum of coordinates, number of inversions), *downward* induction, and the extremal principle (consider a minimal counterexample and derive a contradiction).

11 Recurrence Relations, Catalan Numbers, and Partitions

Theorem 11.1 (Linear Recurrences with Constant Coefficients)**.** *Suppose* $a_n = c_1 a_{n-1} + \cdots + c_k a_{n-k}$ *and the characteristic polynomial* $x^k - c_1 x^{k-1} - \cdots - c_k$ *has distinct roots* $r_1, \ldots, r_k$*. Then* $a_n = \sum_i \alpha_i r_i^n$ *for constants* α_i *determined by initial values. (A root* r *of multiplicity* m *contributes* $(\beta_0 + \beta_1 n + \cdots + \beta_{m-1} n^{m-1}) r^n$*.)*

Proof sketch. Each r_i^n satisfies the recurrence by definition of the characteristic polynomial; solutions form a k-dimensional vector space (a solution is determined by k initial values); the k sequences (r_i^n) are linearly independent (Vandermonde determinant), hence a basis. □

Example 11.2. Count binary strings of length n with no two consecutive 1s.

Solution. Let a_n be the count. A valid string ends in 0 (preceded by any valid string of length $n-1$) or in 01 (preceded by any valid string of length $n-2$): $a_n = a_{n-1} + a_{n-2}$, with $a_1 = 2$, $a_2 = 3$. Thus $a_n = F_{n+2}$ (Fibonacci). □

11.1 Catalan Numbers, Partitions, and Non-Intersecting Paths

Theorem 11.3 (Catalan Numbers)**.** *The* n*-th* ***Catalan number*** $C_n = \frac{1}{n+1}\binom{2n}{n}$ *counts: (1)* ***Dyck paths*** *from* $(0,0)$ *to* (n,n) *with steps* $(1,0)$ *and* $(0,1)$ *never rising above* $y = x$*; (2) balanced strings of* n *pairs of parentheses; (3) triangulations of a convex* $(n+2)$*-gon; and satisfies* $C_{n+1} = \sum_{k=0}^{n} C_k C_{n-k}$ *(*$C_0 = 1$*).*

Proof via André's Reflection Principle. There are $\binom{2n}{n}$ total paths from $(0,0)$ to (n,n). A path is *bad* if it crosses above $y = x$, so it touches $y = x+1$. Reflect the portion of a bad path *after* its first touch of $y = x+1$ across the line $y = x+1$. Since (n,n) reflects to $(n-1, n+1)$, this gives a bijection between bad paths and all paths from $(0,0)$ to $(n-1, n+1)$, of which there are $\binom{2n}{n-1}$. Thus

$$C_n = \binom{2n}{n} - \binom{2n}{n-1} = \frac{1}{n+1}\binom{2n}{n}.$$ □

Theorem 11.4 (Euler's Partition Identity and LGV Lemma)**.**

(a) ***(Euler)*** *For every* $n \geq 1$*, the number of partitions of* n *into* odd *parts equals the number of partitions of* n *into* distinct *parts:* $\prod_{k \geq 1} \frac{1}{1-x^{2k-1}} = \prod_{k \geq 1} \frac{1-x^{2k}}{1-x^k} = \prod_{k \geq 1}(1 + x^k)$.

(b) ***(Lindström–Gessel–Viennot)*** *In a planar DAG where any path* $A_i \to B_j$ *and* $A_{i'} \to B_{j'}$ *(*$i < i', j > j'$*) must cross, the number of vertex-disjoint path systems from* $(A_1, \ldots, A_k)$ *to* $(B_1, \ldots, B_k)$ *equals* $\det\big(e(A_i, B_j)\big)_{i,j=1}^{k}$.

Practice Problem 11.5. Let $a_1 = 1$ and $a_{n+1} = 2a_n + 1$. Find a closed form and prove it.

Solution. Add 1 to both sides: $a_{n+1} + 1 = 2(a_n + 1)$. So $b_n = a_n + 1$ is geometric with $b_1 = 2$: $b_n = 2^n$, hence $a_n = 2^n - 1$. (Induction verifies: $a_{n+1} = 2(2^n - 1) + 1 = 2^{n+1} - 1$.) □

12 The Pigeonhole Principle

Theorem 12.1 (Pigeonhole)**.** *If $n+1$ objects are placed into n boxes, some box contains at least two objects. More generally, if N objects go into n boxes, some box contains at least $\lceil N/n \rceil$ objects.*

Proof. If every box had at most $\lceil N/n \rceil - 1 < N/n$ objects, the total would be less than N. □

Example 12.2. Among any $n+1$ integers chosen from $\{1, 2, \ldots, 2n\}$, one divides another.

Solution. Write each chosen integer as $2^a \cdot m$ with m odd. There are only n odd numbers in $\{1, \ldots, 2n\}$, so two chosen integers share the same odd part m: say $2^a m$ and $2^b m$ with $a < b$. Then the first divides the second. □

Practice Problem 12.3. Prove that among any n integers, some nonempty subset has sum divisible by n.

Solution. Let the integers be $a_1, \ldots, a_n$ and consider the prefix sums $s_0 = 0,\ s_1 = a_1,\ \ldots,\ s_n = a_1 + \cdots + a_n$. These are $n+1$ numbers and only n residues mod n, so $s_i \equiv s_j \pmod{n}$ for some $i < j$; then $a_{i+1} + \cdots + a_j = s_j - s_i$ is divisible by n. □

13 Bijective Counting, Inclusion–Exclusion, and Posets

Definition 13.1. For finite sets, $|A \cup B| = |A| + |B| - |A \cap B|$; more generally, **inclusion–exclusion**:

$$\Big|\bigcup_{i=1}^{n} A_i\Big| = \sum_{\emptyset \neq S \subseteq [n]} (-1)^{|S|+1} \Big|\bigcap_{i \in S} A_i\Big|.$$

Proof. Fix an element x lying in exactly $k \geq 1$ of the sets. Its contribution to the right-hand side is $\sum_{j=1}^{k} (-1)^{j+1} \binom{k}{j} = 1 - (1-1)^k = 1$, matching the left side. □

Bijections and double counting. If a bijection exists between two finite sets, they have equal size; counting one quantity in two ways yields an identity. Injections give $|A| \leq |B|$.

Example 13.2 (Derangements)**.** The number of permutations of $[n]$ with no fixed point is $D_n = n! \sum_{k=0}^{n} \frac{(-1)^k}{k!}$.

Solution. Let A_i be the set of permutations fixing i. By inclusion–exclusion,

$$|\bigcup A_i| = \sum_{k} (-1)^{k+1} \binom{n}{k} (n-k)!$$

so $D_n = n! - \sum_{k \geq 1} (-1)^{k+1} \frac{n!}{k!} = n! \sum_{k=0}^{n} \frac{(-1)^k}{k!}$. □

13.1 Sperner's Theorem and Dilworth's Theorem [Advanced]

Theorem 13.3 (LYM Inequality, Sperner, Mirsky, and Dilworth)**.**

(a) ***(LYM & Sperner)*** *Let $\mathcal{F}$ be an antichain of subsets of $[n]$ (no set contains another). Then*

$$\sum_{A \in \mathcal{F}} \frac{1}{\binom{n}{|A|}} \leq 1, \qquad \textit{and hence} \qquad |\mathcal{F}| \leq \binom{n}{\lfloor n/2 \rfloor}.$$

(b) ***(Mirsky & Dilworth)*** *In any finite poset $(P, \leq)$, the maximum size of a chain equals the minimum number of antichains needed to partition P, and the maximum size of an antichain equals the minimum number of chains needed to partition P.*

Proof. (a) Among all $n!$ maximal chains $\emptyset = C_0 \subset C_1 \subset \cdots \subset C_n = [n]$, each $A \in \mathcal{F}$ belongs to $|A|!(n-|A|)!$ chains, and no chain contains two members of $\mathcal{F}$. Thus $\sum_{A\in\mathcal{F}} |A|!(n-|A|)! \leq n!$, which divides by $n!$ to give LYM; using $\binom{n}{|A|} \leq \binom{n}{\lfloor n/2 \rfloor}$ gives Sperner's bound. (b) For Mirsky, let $h(x)$ be the length of the longest chain ending at x; then $x < y \implies h(x) < h(y)$, so the level sets $A_k = \{x : h(x) = k\}$ partition P into $\max h$ antichains. □

Practice Problem 13.4 (Double Counting and Erdős–Szekeres)**.** (a) At a party each pair of people are friends or strangers. Prove that the number of people with an odd number of friends is even. (b) Prove that any sequence of $mn+1$ real numbers has a nondecreasing subsequence of length $m+1$ or a decreasing subsequence of length $n+1$.

Solution. (a) Count pairs (person, friend of that person): the total is $\sum_v \deg(v) = 2E$, twice the number of friendships, hence even. A sum of integers is even only if the number of odd summands is even. (b) Map each $i \in \{1, \ldots, mn+1\}$ to (ℓ_i, d_i), the lengths of the longest nondecreasing subsequence ending at a_i and longest decreasing starting at a_i. For $i < j$, $a_i \leq a_j \implies \ell_j \geq \ell_i + 1$ and $a_i > a_j \implies d_i \geq d_j + 1$, so the map is injective and cannot fit in $\{1, \ldots, m\} \times \{1, \ldots, n\}$. □

14 Invariants, Monovariants, and Coloring Arguments

Definition 14.1. Under a set of allowed moves on a state space:

- An **invariant** $I(S)$ is unchanged by every move ($I(S') = I(S)$). If $I(S_{\text{start}}) \neq I(S_{\text{target}})$, then S_{target} is unreachable.
- A **monovariant** $M(S)$ strictly increases (or strictly decreases) on every move. If M takes values in a finite or discrete bounded set, the process must terminate.

Example 14.2 (Checkerboard Coloring)**.** An 8×8 board with two opposite corners removed cannot be tiled by 31 dominoes of size 1×2.

Solution. Each 1×2 domino covers one black and one white square on a standard chessboard coloring, whereas the mutilated board has 32 squares of one color and 30 of the other. □

Practice Problem 14.3 (Grid Infection Monovariant)**.** Initially, k unit squares of an $n \times n$ grid are infected. At each step, any uninfected square sharing an edge with at least 2 infected squares becomes infected. Prove that if the entire $n \times n$ grid becomes infected, $k \geq n$.

Solution. Let P be the perimeter of the infected region. When a new square is infected, it shares $c \geq 2$ edges with already-infected squares, so those c edges disappear from the boundary while $4-c$ edges are added. Thus $\Delta P = (4-c) - c = 4 - 2c \leq 0$: the perimeter P never increases! Initially $P_{\text{init}} \leq 4k$, and at the end $P_{\text{final}} = 4n$. Hence $4n \leq 4k \implies k \geq n$. □

15 The Extremal Principle and Combinatorial Games

Theorem 15.1 (Sylvester–Gallai Theorem)**.** *Given a finite set S of at least 3 non-collinear points in the plane, there exists a line passing through* exactly two *points of S.*

Proof (Kelly). Among all pairs (P, ℓ) where ℓ is a line through at least two points of S and $P \in S \setminus \ell$, choose one minimizing the perpendicular distance $d(P, \ell) > 0$. If ℓ contained ≥ 3 points of S, at least two of them (B and C) would lie on the same side of the foot H of P on ℓ, with B between H and C (possibly $B = H$). Then the distance from B to line PC is strictly less than $PH = d(P, \ell)$, contradicting minimality. □

Theorem 15.2 (Combinatorial Games, Nim, and Sprague–Grundy)**.** *In a finite, impartial, normal-play game (last player to move wins):*

- *A position is a **losing (P) position** if all moves from it lead to N-positions, and a **winning (N) position** if some move leads to a P-position.*
- ***(Bouton / Nim)** In Nim with pile sizes $(n_1, \ldots, n_k)$, a state is a P-position iff the XOR sum $n_1 \oplus \cdots \oplus n_k = 0$.*
- ***(Sprague–Grundy)** Every impartial game position G is equivalent to a single Nim pile of size $g(G) = \operatorname{mex}\{g(G') : G \to G'\}$, and independent games add via $\oplus$.*

Proof for Nim. Terminal $(0, \ldots, 0)$ has XOR sum 0. Changing a single n_i to $n_i' < n_i$ alters the highest bit where n_i, n_i' differ, so XOR sum 0 always moves to XOR sum $s \neq 0$. Conversely, if $s = n_1 \oplus \cdots \oplus n_k \neq 0$, pick n_i having a 1 in the highest bit of s; then $n_i' = n_i \oplus s < n_i$, and replacing $n_i \to n_i'$ gives XOR sum 0. □

Practice Problem 15.3 (Strategy Stealing in Chomp)**.** In Chomp on an $m \times n$ chocolate bar ($mn > 1$), players take turns choosing a remaining square (i, j) and eating all (i', j') with $i' \geq i, j' \geq j$; eating $(1, 1)$ loses. Prove Player 1 has a winning strategy.

Solution. Suppose Player 2 had a winning strategy. Let Player 1 eat only the corner (m, n) on turn 1. Player 2's winning reply eats from some (a, b). But eating (a, b) on turn 1 would have produced the exact same board! Thus Player 1 can play (a, b) on turn 1 and steal Player 2's winning strategy, contradiction. □

16 Probability and the Probabilistic Method [Useful]

Definition 16.1. A finite probability space assigns weights $p(\omega) \geq 0$, $\sum p(\omega) = 1$, to outcomes. For an event A, $P(A) = \sum_{\omega \in A} p(\omega)$. A random variable X has **expected value** $E[X] = \sum_\omega p(\omega) X(\omega)$.

Theorem 16.2 (Linearity of Expectation, Alteration, and Lovász Local Lemma)**.**

*(a) **(Linearity)** For any random variables (independent or not!), $E[X + Y] = E[X] + E[Y]$ and $E[cX] = cE[X]$.*

*(b) **(Existence & Union Bound)** Some outcome has $X(\omega) \geq E[X]$ and some has $X(\omega) \leq E[X]$. If $\sum P(A_i) < 1$, then $P(\bigcap \overline{A_i}) > 0$.*

*(c) **(Alteration / Deletion Method)** If a random sample S has expected size $E[X]$ and $E[Y]$ bad substructures, deleting one element per bad substructure leaves a clean subset of size $\geq E[X] - E[Y]$.*

*(d) **(Lovász Local Lemma)** If bad events A_i satisfy $P(A_i) \leq p$, each A_i is mutually independent of all but at most d other A_j, and $ep(d + 1) \leq 1$, then $P(\bigcap \overline{A_i}) > 0$.*

Example 16.3. A random permutation of $[n]$ has, on average, 1 fixed point ($E[\sum X_i] = n \cdot \frac{1}{n} = 1$).

Practice Problem 16.4 (Caro–Wei Theorem and Alteration Bound)**.** Let $G = (V, E)$ be a graph with n vertices, m edges, and average degree $d = 2m/n \geq 1$. (a) Prove G has an independent set of size at least $\sum_{v \in V} \frac{1}{\deg(v)+1}$. (b) Use the alteration method to prove G has an independent set of size at least $\frac{n}{2d}$.

Solution. (a) Pick a uniformly random permutation π of V and let I be the set of vertices v that appear in π before all of their neighbors. No edge can have both endpoints in I, so I is independent. Each v is first among $\{v\} \cup N(v)$ with probability $\frac{1}{\deg(v)+1}$, so $E[|I|] = \sum_{v \in V} \frac{1}{\deg(v)+1}$. (b) Pick each vertex independently with probability $p = 1/d$, and delete one endpoint from each edge with both endpoints chosen. The expected remaining size is $np - mp^2 = np - \frac{nd}{2}p^2 = \frac{n}{2d}$. □

17 Graph Theory

Definition 17.1. A **graph** $G = (V, E)$ consists of vertices and edges. The **degree** $\deg(v)$ is the number of edges at v. A **tree** is a connected graph with no cycles.

Theorem 17.2 (Handshake Lemma and Trees)**.** $\sum_{v \in V} \deg(v) = 2|E|$. *A tree on n vertices has $n - 1$ edges; every connected graph on n vertices has at least $n - 1$ edges.*

Proof. Each edge contributes 2 to $\sum \deg(v)$. A tree has a leaf (endpoint of a maximal path); remove it and induct. □

17.1 Eulerian, Hamiltonian, and Planar Graphs

Theorem 17.3 (Eulerian Circuits, Dirac's Theorem, and Euler's Planar Formula)**.**

(a) ***(Euler)*** *A connected graph has an Eulerian circuit iff every vertex has even degree.*

(b) ***(Dirac)*** *If $n \geq 3$ and $\deg(v) \geq n/2$ for all v, then G has a Hamiltonian cycle.*

(c) ***(Euler's Planar Formula)*** *Any connected plane graph satisfies $V - E + F = 2$. For $V \geq 3$, $3F \leq 2E \implies E \leq 3V - 6$ ($E \leq 2V - 4$ if triangle-free), so every planar graph has a vertex of degree ≤ 5.*

Proof of (c). For a tree, $E = V - 1$ and $F = 1$, so $V - E + F = 2$. Deleting an edge on a cycle reduces E and F by 1 each, preserving $V - E + F$. □

Practice Problem 17.4 (Hamiltonian Path in a Tournament)**.** Prove that every tournament on n players (every pair plays, one wins) contains a Hamiltonian path $p_1 \to p_2 \to \cdots \to p_n$.

Solution. Induct on n. Given $p_1 \to \cdots \to p_{n-1}$ and a new player v: if v beats p_1, prepend v; if p_{n-1} beats v, append v. Otherwise let i be the first index with v beating p_{i+1}; then p_i beats v, so insert v between p_i and p_{i+1}. □

17.2 Hall's Marriage Theorem, König's Theorem, and Stable Matching [Advanced]

Theorem 17.5 (Hall, König, and Gale–Shapley)**.**

(a) ***(Hall's Marriage Theorem)*** *A bipartite graph (A, B) has a matching covering A iff $|N(S)| \geq |S|$ for every $S \subseteq A$.*

(b) ***(König)*** *In any bipartite graph, max matching size = min vertex cover size.*

(c) ***(Gale–Shapley)*** *Given n men and n women with strict preferences, the propose-and-reject algorithm always terminates in a stable matching.*

Proof of Hall. Necessity is clear. For sufficiency, induct on $|A|$. If $|N(S)| \geq |S|+1$ for all nonempty proper $S \subsetneq A$, match any edge ab and induct on $A \setminus \{a\}$. Otherwise some proper S_0 has $|N(S_0)| = |S_0|$. Match S_0 onto $N(S_0)$ by induction; for $T \subseteq A \setminus S_0$, $|N(T) \setminus N(S_0)| = |N(T \cup S_0)| - |N(S_0)| \geq |T \cup S_0| - |S_0| = |T|$, so Hall's condition holds on $(A \setminus S_0, B \setminus N(S_0))$. □

Practice Problem 17.6. A deck of mn cards (n values, each repeated m times) is dealt into n piles of m cards. Prove one can pick one card from each pile so that all n values appear.

Solution. Form a bipartite graph between piles and values. Any k piles hold km cards; since each value appears only m times in total, at least $\lceil km/m \rceil = k$ distinct values appear in those k piles. By Hall's theorem, a perfect matching exists. □

17.3 Extremal Graph Theory and Ramsey Theory [Advanced]

Theorem 17.7 (Mantel, Turán, Kővári–Sós–Turán, and Ramsey)**.**

(a) ***(Mantel)*** *A triangle-free graph on n vertices has at most $\lfloor n^2/4 \rfloor$ edges.*

(b) ***(Turán)*** *A K_{r+1}-free graph on n vertices has at most $\left(1 - \frac{1}{r}\right)\frac{n^2}{2}$ edges (achieved by the balanced r-partite Turán graph).*

(c) ***(Kővári–Sós–Turán)*** *A C_4-free graph on n vertices satisfies $|E| \leq \frac{n}{4}(1 + \sqrt{4n-3}) = O(n^{3/2})$.*

(d) ***(Ramsey)*** *Every 2-coloring of K_N with $N \geq R(s,t)$ has a red K_s or blue K_t, where $R(s,t) \leq R(s-1,t) + R(s,t-1) \leq \binom{s+t-2}{s-1}$ and $R(3,3) = 6$.*

Proof. (a) For each edge uv, $\deg u + \deg v \leq n$; summing over edges gives $\sum_v (\deg v)^2 = \sum_{uv}(\deg u + \deg v) \leq n|E|$. By Cauchy–Schwarz, $\frac{4|E|^2}{n} \leq \sum(\deg v)^2 \leq n|E| \implies |E| \leq n^2/4$. (c) Count 2-edge paths $(v, \{u,w\})$: since G is C_4-free, each $\{u,w\}$ has at most 1 common neighbor, so $n\binom{2|E|/n}{2} \leq \sum_v \binom{\deg v}{2} \leq \binom{n}{2}$. (d) A vertex in $K_{R(s-1,t)+R(s,t-1)}$ has either $\geq R(s-1,t)$ red neighbors or $\geq R(s,t-1)$ blue neighbors; induct. □

18 Algebraic Combinatorics and Combinatorial Geometry [Advanced]

18.1 Linear Algebra (Oddtown), Nullstellensatz, and Burnside's Lemma

Theorem 18.1 (Oddtown, Combinatorial Nullstellensatz, and Burnside's Lemma)**.**

(a) ***(Oddtown)*** *If $A_1, \ldots, A_m \subseteq [n]$ have $|A_i|$ odd and $|A_i \cap A_j|$ even ($i \neq j$), then $m \leq n$.* Proof: *Incidence vectors $v_i \in \mathbb{F}_2^n$ satisfy $v_i \cdot v_j = \delta_{ij} \pmod 2$, so they are linearly independent in $\mathbb{F}_2^n$.*

(b) ***(Alon's Combinatorial Nullstellensatz)*** *If* $\deg P = \sum d_i$ *and* $[x_1^{d_1} \cdots x_n^{d_n}]P \neq 0$*, then for any* $S_i \subseteq F$ *with* $|S_i| > d_i$*, there exists* $s \in S_1 \times \cdots \times S_n$ *with* $P(s) \neq 0$ *(reduce* P *modulo* $\prod_{s \in S_i}(x_i - s)$*; the leading coefficient survives so the remainder is nonzero).*

(c) ***(Cauchy–Davenport)*** *For nonempty* $A, B \subseteq \mathbb{Z}/p\mathbb{Z}$*,* $|A + B| \geq \min(p, |A| + |B| - 1)$ *(apply Nullstellensatz to* $\prod_{c \in C}(x + y - c)$ *where* $[x^{|A|-1}y^{|B|-1}] = \binom{|A|+|B|-2}{|A|-1} \not\equiv 0 \pmod{p}$*).*

(d) ***(Burnside's Lemma)*** *The number of orbits of a finite group* G *acting on* X *is* $|X/G| = \frac{1}{|G|}\sum_{g \in G} |X^g|$ *(double count* $\{(g, x) : g \cdot x = x\}$*).*

18.2 Convex Hulls, Pick's Theorem, and Helly's Theorem

Theorem 18.2 (Convex Hulls, Pick's Theorem, and Helly's Theorem)**.**

(a) *For finite* $S \subset \mathbb{R}^2$*, the* ***convex hull*** *is the set of convex combinations* $\sum \lambda_i p_i$ *(*$\lambda_i \geq 0, \sum \lambda_i = 1$*); its vertices lie in* S*.*

(b) ***(Pick's Theorem)*** *The area of a simple lattice polygon is* $A = I + \frac{B}{2} - 1$*.*

(c) ***(Helly's Theorem)*** *If every* $d+1$ *sets in a finite family of convex sets in* $\mathbb{R}^d$ *have a common point, then all sets in the family have a common point.*

Practice Problem 18.3. Given $2n$ points in general position (no 3 collinear), n red and n blue, prove they can be split into n red–blue pairs whose connecting segments are pairwise non-crossing.

Solution. Among all perfect red–blue pairings, choose one minimizing the total length of the n segments. If two segments R_1B_1 and R_2B_2 crossed at X, then by the triangle inequality $R_1B_2 + R_2B_1 < (R_1X + XB_2) + (R_2X + XB_1) = R_1B_1 + R_2B_2$, so swapping partners strictly decreases total length — contradiction. □

Part III
Geometry

Throughout, ABC is a triangle with side lengths $a = BC$, $b = CA$, $c = AB$, semiperimeter $s = \frac{a+b+c}{2}$, circumradius R, inradius r, and area $[ABC]$.

19 Triangle Centers, the Nine-Point Circle, and Fact 5

Definition 19.1.

- **Centroid** G: intersection of medians; divides each median $2:1$ from the vertex.
- **Incenter** I: intersection of internal angle bisectors; center of the inscribed circle, equidistant from the sides.
- **Circumcenter** O: intersection of perpendicular bisectors; center of the circle through A, B, C.
- **Orthocenter** H: intersection of the altitudes.
- **Excenter** I_A: intersection of the internal bisector from A and external bisectors from B, C; center of the excircle opposite A.

Theorem 19.2 (Concurrency of Medians)**.** *The three medians meet at a point G dividing each in ratio $2:1$.*

Proof. With position vectors, let $G = \frac{A+B+C}{3}$. The midpoint of BC is $M = \frac{B+C}{2}$, and $G = A + \frac{2}{3}(M - A)$ lies on median AM at ratio $2:1$; by symmetry G lies on all three medians. □

Theorem 19.3 (Euler Line)**.** *O, G, H are collinear with $\vec{OH} = 3\,\vec{OG}$, equivalently $H = A+B+C$ when O is the origin.*

Proof. Take O as origin, so $|A| = |B| = |C| = R$. Let $H' = A + B + C$. Then $H' - A = B + C$ is perpendicular to $B - C$ since $(B + C) \cdot (B - C) = |B|^2 - |C|^2 = 0$; thus $AH' \perp BC$, i.e. H' lies on the altitude from A, and by symmetry on all altitudes, so $H' = H$. Since $G = \frac{A+B+C}{3} = \frac{H}{3}$, the collinearity and ratio follow. □

Theorem 19.4 (Nine-Point Circle)**.** *The midpoints of the sides, the feet of the altitudes, and the midpoints of AH, BH, CH all lie on one circle with center N (the midpoint of OH) and radius $R/2$.*

Proof. Consider the homothety h centered at H with ratio $\frac{1}{2}$. It maps the circumcircle to a circle of radius $R/2$ centered at the midpoint N of OH. It maps A, B, C to the midpoints of HA, HB, HC: three of the nine points. Next, the reflection of H over the midpoint M_A of BC is the antipode A' of A on the circumcircle (indeed $BHCA'$ is a parallelogram: $BH \parallel CA'$ as both are perpendicular to AC, and $CH \parallel BA'$ likewise); hence $h(A') = M_A$, putting the three side midpoints on the circle. Finally the reflection of H over the foot of the altitude from A lies on the circumcircle (reflecting H across BC gives a point H_A with $\angle BH_AC = \angle BHC = 180^\circ - \angle A$, so H_A is concyclic with A, B, C), hence $h(H_A)$ is the foot itself, and the three feet lie on the circle too. □

19.1 The Incenter–Excenter Lemma and Intouch/Extouch Symmetry

Theorem 19.5 (The Incenter–Excenter Lemma ("Fact 5"))**.** *Let M be the midpoint of arc BC of (ABC) not containing A. Then M is the center of a circle passing through B, I, C, I_A:*

$$MB = MC = MI = MI_A.$$

Proof. Since M bisects arc BC, line AM bisects $\angle A$ (so A, I, M, I_A are collinear) and $MB = MC$. By angle chasing, $\angle MBI = \angle MBC + \angle CBI = \angle MAC + \frac{\angle B}{2} = \frac{\angle A+\angle B}{2}$, while by the exterior angle theorem on $\triangle ABI$, $\angle BIM = \frac{\angle A+\angle B}{2}$. Thus $MB = MI$. Furthermore, internal and external bisectors at B and C are perpendicular, so $\angle IBI_A = \angle ICI_A = 90^\circ$. Hence B, I, C, I_A lie on the circle with diameter II_A, whose center on II_A must be M since $MB = MI = MC$. □

Proposition 19.6 (Tangent Lengths and Euler's Formula $OI^2 = R(R - 2r)$)**.**

(a) If the incircle touches BC, CA, AB at D, E, F and the A-excircle touches BC at D_a, then $AE = AF = s - a$, $BD = BF = s - b$, $CD = CE = s - c$, and $BD = CD_a = s - b$ (so D and D_a are symmetric about the midpoint of BC).

(b) $OI^2 = R^2 - 2Rr$, so $R \geq 2r$.

Proof of (b). Let AI meet (ABC) at M, and let M' be the antipode of M on (ABC) and D the foot from I to AB ($ID = r$). By power of a point and Fact 5, $R^2 - OI^2 = AI \cdot IM = AI \cdot BM$. Since $\triangle AID \sim \triangle M'MB$, $\frac{AI}{2R} = \frac{r}{BM} \implies AI \cdot BM = 2Rr$. □

Practice Problem 19.7. Prove that the reflection of the orthocenter H over side BC lies on the circumcircle of ABC.

Solution. In the quadrilateral formed by the altitudes from B and C, $\angle BHC = 180^\circ - \angle A$. Reflecting H over BC to H' preserves the angle, so $\angle BH'C = \angle BHC = 180^\circ - \angle BAC$, meaning $ABH'C$ is cyclic. □

20 Angle Chasing, Directed Angles, and Cyclic Quadrilaterals

Theorem 20.1 (Inscribed Angle Theorem)**.** *An inscribed angle equals half the central angle subtending the same arc. Consequently: angles subtending the same arc are equal, and an angle in a semicircle is right.*

Proof. If one side of the inscribed angle at P is a diameter PQ', $\triangle OPQ$ is isosceles ($OP = OQ = R$), so the central angle $\angle Q'OQ = 2\angle OPQ$. The general case adds or subtracts two such angles. □

Directed Angles Modulo 180°**.** Write $\measuredangle(\ell_1, \ell_2)$ for the counterclockwise angle from line ℓ_1 to line ℓ_2 modulo 180°, and $\measuredangle ABC := \measuredangle(BA, BC)$. Then $\measuredangle ABC = -\measuredangle CBA$ and $\measuredangle AXP + \measuredangle PXB = \measuredangle AXB$.

Theorem 20.2 (Cyclic Quadrilateral Criteria)**.** *For a convex quadrilateral $ABCD$, the following are equivalent: (i) $ABCD$ is cyclic; (ii) $\angle ABD = \angle ACD$; (iii) $\angle ABC + \angle CDA = 180^\circ$; (iv) for any four non-collinear points, $\measuredangle ABD = \measuredangle ACD$.*

Proof. (i)⇒(ii)–(iv) by the inscribed angle theorem. Conversely, if $\measuredangle ABD = \measuredangle ACD$ and line AC meets circle (ABD) again at C', then $\measuredangle AC'D = \measuredangle ABD = \measuredangle ACD$, so lines $C'D$ and CD are parallel and share D, forcing $C' = C$. □

Theorem 20.3 (Ptolemy's Theorem)**.** *For a cyclic quadrilateral $ABCD$: $AC \cdot BD = AB \cdot CD + AD \cdot BC$. In general (Ptolemy's inequality) $AC \cdot BD \le AB \cdot CD + AD \cdot BC$ with equality iff $ABCD$ is cyclic in this order.*

Proof. Choose E on diagonal AC with $\angle ABE = \angle DBC$. Since $\angle BAE = \angle BDC$, triangles ABE and DBC are similar, so $AE \cdot BD = AB \cdot CD$. Also $\angle EBC = \angle ABD$ and $\angle BCE = \angle BDA$, so $\triangle EBC \sim \triangle ABD$, giving $EC \cdot BD = AD \cdot BC$. Adding gives $(AE + EC) \cdot BD = AC \cdot BD = AB \cdot CD + AD \cdot BC$. □

Theorem 20.4 (Simson Line and Miquel's Theorem)**.**

(a) ***(Simson Line)*** *The feet X, Y, Z of the perpendiculars from P to BC, CA, AB are collinear iff P lies on (ABC).*

(b) ***(Miquel's Theorem)*** *If D, E, F lie on lines BC, CA, AB, the circles (AEF), (BFD), and (CDE) concur at a point M.*

Proof. (a) Since P, Y, Z, A and P, X, Z, B are concyclic, $\measuredangle YZX = \measuredangle YZP + \measuredangle PZX = \measuredangle YAP + \measuredangle PBX = \measuredangle CAP - \measuredangle CBP$, which is 0 (mod 180°) iff $P \in (ABC)$. (b) If (AEF) and (BFD) meet at $M \neq F$, then $\measuredangle DME = \measuredangle DMF + \measuredangle FME = \measuredangle DBF + \measuredangle FAE = \measuredangle DCE$, so $M \in (CDE)$. □

Practice Problem 20.5. Point P lies on the arc BC of the circumcircle of equilateral triangle ABC (not containing A). Prove $PA = PB + PC$.

Solution. Apply Ptolemy to cyclic quadrilateral $ABPC$: $PA \cdot BC = PB \cdot AC + PC \cdot AB$. Since $AB = BC = CA$, divide through to get $PA = PB + PC$. □

21 Similar Triangles and Metric Relations

Theorem 21.1 (Similarity Criteria and Stewart's Theorem)**.**

(a) *Triangles are similar under* ***AA***, ***SAS***, *or* ***SSS****; areas scale as the square of the similarity ratio.*

(b) ***(Stewart's Theorem)*** *For $D \in BC$ with $BD = m, DC = n, m+n = a$, and $AD = d$:*

$$man + dad = bmb + cnc \quad \big(a(mn+d^2) = b^2m + c^2n\big).$$

Consequently, median $m_a^2 = \frac{2b^2+2c^2-a^2}{4}$ and internal bisector $\ell_a^2 = bc\big(1 - \frac{a^2}{(b+c)^2}\big)$.

(c) ***(Heron & Brahmagupta)*** *$[ABC] = \sqrt{s(s-a)(s-b)(s-c)}$, and a cyclic quad with sides a, b, c, d has area $\sqrt{(s-a)(s-b)(s-c)(s-d)}$.*

Proof of Stewart. Let $\theta = \angle ADB$; by the Law of Cosines on $\triangle ABD$ and $\triangle ACD$, $c^2 = m^2 + d^2 - 2md\cos\theta$ and $b^2 = n^2 + d^2 + 2nd\cos\theta$. Multiply the first by n, the second by m, and add. □

Example 21.2. In $\triangle ABC$, $DE \parallel BC$ with $D \in AB, E \in AC$. If $AD = 3, DB = 2, BC = 10$, then $\triangle ADE \sim \triangle ABC$ with ratio $\frac{3}{5}$, so $DE = 6$.

Practice Problem 21.3. In right triangle ABC ($\angle C = 90^\circ$) with altitude $CH \perp AB$, prove $CH^2 = AH \cdot BH$ and $AC^2 = AH \cdot AB$.

Solution. Since $\angle ACH = 90^\circ - \angle A = \angle B$, $\triangle ACH \sim \triangle CBH \sim \triangle ABC$ (AA), giving $\frac{CH}{BH} = \frac{AH}{CH}$ and $\frac{AC}{AB} = \frac{AH}{AC}$. □

22 Power of a Point, Radical Axes, and Apollonius Circles

Definition 22.1. The **power** of point P with respect to circle $\omega(O, r)$ is $\mathrm{pow}(P, \omega) = OP^2 - r^2$.

Theorem 22.2 (Power of a Point, Radical Center, and Circle of Apollonius)**.**

(a) *If a line through P meets ω at X, Y, then $\overrightarrow{PX} \cdot \overrightarrow{PY} = OP^2 - r^2$ (and $= PT^2$ for tangent PT). Conversely, if $\overrightarrow{PX} \cdot \overrightarrow{PY} = \overrightarrow{PZ} \cdot \overrightarrow{PW}$ on two lines through P, then X, Y, Z, W are concyclic.*

(b) ***(Radical Axis & Coaxial Pencils)*** *The locus of points with equal power to non-concentric ω_1, ω_2 is a line $\perp O_1O_2$ (the* ***radical axis****). For three circles, the pairwise radical axes concur at the* ***radical center****. A disjoint pencil $\omega_1 + \lambda\omega_2 = 0$ has two radius-0 limiting points L_1, L_2; inverting at L_1 maps the pencil to concentric circles.*

(c) ***(Circle of Apollonius)*** *For distinct A, B and $k > 0$ ($k \neq 1$), the locus $\{P : PA/PB = k\}$ is the circle with diameter DD', where $D, D' \in AB$ are the internal and external angle bisector feet ($\frac{DA}{DB} = \frac{D'A}{D'B} = k$), since PD, PD' bisect $\angle APB \iff \angle DPD' = 90^\circ$.*

Proof of (a). With M the midpoint of XY, $OM \perp XY$ and $PX \cdot PY = PM^2 - MX^2 = (OP^2 - OM^2) - (r^2 - OM^2) = OP^2 - r^2$. □

Practice Problem 22.3. Two circles intersect at A and B. A line through B meets the circles again at C and D. Tangents to the circles at C and D meet at P. Prove that P, A, C, D are concyclic.

Solution. By tangent–chord angles, $\angle PCB = \angle CAB$ and $\angle PDB = \angle DAB$. Thus $\angle CPD = 180^\circ - \angle PCD - \angle PDC = 180^\circ - \angle CAB - \angle DAB = 180^\circ - \angle CAD$, so $PCAD$ is cyclic. □

23 Homothety, Spiral Similarity, and Complete Quadrilaterals

Definition 23.1. A **homothety** $h(O,k)$ maps $P \mapsto O + k(P-O)$. A **spiral similarity** centered at O is a rotation composed with a homothety at O: $z \mapsto \alpha(z-o)+o$.

Theorem 23.2 (Monge, Spiral Similarity Lemma, and Complete Quadrilaterals)**.**

(a) ***(Monge's Theorem)*** *For three circles of distinct radii, the three external homothety centers are collinear.*

(b) ***(Spiral Similarity Lemma)*** *If lines AC and BD meet at X, the second intersection P of circles (XAB) and (XCD) is the center of the spiral similarity sending $A \mapsto B, C \mapsto D$ and the spiral similarity sending $A \mapsto C, B \mapsto D$: $\triangle PAB \sim \triangle PCD$ and $\triangle PAC \sim \triangle PBD$.*

(c) ***(Complete Quadrilateral)*** *Given four lines forming four triangles, their four circumcircles concur at the Miquel point M (the spiral center), M and the four circumcenters are concyclic, M has a common Simson line, the four orthocenters are collinear, and the midpoints of the three diagonals are collinear (**Newton–Gauss line**).*

Proof of (b). In cyclic quads $PXAB$ and $PXCD$: $\measuredangle PAB = \measuredangle PXB = \measuredangle PXD = \measuredangle PCD$ and $\measuredangle PBA = \measuredangle PXA = \measuredangle PXC = \measuredangle PDC$, so $\triangle PAB \sim \triangle PCD$, which also implies $\triangle PAC \sim \triangle PBD$. □

Practice Problem 23.3. Circles ω_1 and ω_2 are internally tangent at T, with ω_1 inside. A chord AB of ω_2 is tangent to ω_1 at P. Prove that line TP bisects the arc AB of ω_2 not containing T.

Solution. The homothety h at T taking $\omega_1 \to \omega_2$ maps P to $M = h(P) \in \omega_2$, and maps tangent AB of ω_1 to the tangent of ω_2 at M, which is parallel to AB. Thus M is the midpoint of arc AB. □

24 Ceva, Menelaus, Isogonal Conjugates, and Symmedians

Theorem 24.1 (Ceva and Menelaus)**.** *For D, E, F on lines BC, CA, AB:*

- ***(Ceva)*** *AD, BE, CF concur iff $\dfrac{BD}{DC}\cdot\dfrac{CE}{EA}\cdot\dfrac{AF}{FB} = 1$ (signed ratios; proved via $\frac{BD}{DC} = \frac{[ABP]}{[ACP]}$).*
- ***(Menelaus)*** *D, E, F are collinear iff $\dfrac{BD}{DC}\cdot\dfrac{CE}{EA}\cdot\dfrac{AF}{FB} = -1$ (signed ratios; proved by dropping perpendiculars to DEF).*

Theorem 24.2 (Isogonal Conjugates and Symmedians)**.**

(a) *Reflecting concurrent cevians AP, BP, CP across the angle bisectors yields concurrent cevians meeting at the **isogonal conjugate** P^* (by Trig Ceva; e.g. $O^* = H$).*

(b) *The isogonal conjugate of median AM_A is the **A-symmedian** (concurring at $K = G^*$). If the tangents to (ABC) at B, C meet at X and AX meets BC at D and (ABC) at T, then AD is the A-symmedian, $\frac{BD}{DC} = \frac{c^2}{b^2}$, and $ABTC$ is a **harmonic quadrilateral** ($AB\cdot CT = AC\cdot BT$).*

Proof of (b). Using $XB = XC$ and $\angle ABX = 180^\circ - C, \angle ACX = 180^\circ - B$, the Law of Sines on $\triangle ABX$ and $\triangle ACX$ gives $\frac{\sin\angle BAX}{\sin\angle CAX} = \frac{\sin C}{\sin B} = \frac{c}{b} = \frac{\sin\angle M_A AC}{\sin\angle BAM_A}$, so AX is the A-symmedian. □

Practice Problem 24.3. Prove using Ceva that the internal angle bisectors of a triangle are concurrent.

Solution. By the Angle Bisector Theorem, $\frac{BD}{DC} = \frac{c}{b}$, $\frac{CE}{EA} = \frac{a}{c}$, $\frac{AF}{FB} = \frac{b}{a}$; their product is 1. □

25 Trigonometric and Coordinate Methods [Useful]

Theorem 25.1 (Law of Sines, Law of Cosines, and Trig Ceva)**.** $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$; $a^2 = b^2 + c^2 - 2bc\cos A$; $[ABC] = \frac{1}{2}bc\sin A = \frac{abc}{4R} = rs$. *Cevians* AD, BE, CF *concur iff* $\frac{\sin\angle BAD}{\sin\angle DAC}\frac{\sin\angle CBE}{\sin\angle EBA}\frac{\sin\angle ACF}{\sin\angle FCB} = 1$.

- **Cartesian & Shoelace:** $[P_1\cdots P_n] = \frac{1}{2}\left|\sum_i (x_i y_{i+1} - x_{i+1}y_i)\right|$.
- **Complex Numbers on Unit Circle ($|a| = |b| = |c| = 1$, $\bar{a} = 1/a$):** $o = 0$, $g = \frac{a+b+c}{3}$, $h = a + b + c$; chord ab: $z + ab\bar{z} = a + b$; tangent at a: $z + a^2\bar{z} = 2a$; foot from p to ab: $\frac{1}{2}(a + b + p - ab\bar{p})$.
- **Barycentric Coordinates $P = (x : y : z)$:** $A = (1 : 0 : 0)$, $G = (1 : 1 : 1)$, $I = (a : b : c)$, $K = (a^2 : b^2 : c^2)$; circumcircle is $a^2yz + b^2zx + c^2xy = 0$.

Practice Problem 25.2. With circumcenter $O = 0$ and $|a| = |b| = |c| = 1$, prove the reflection of O across AB is $a + b$, and that it is the circumcenter of $\triangle HAB$.

Solution. The midpoint of AB is $\frac{a+b}{2}$, so reflecting 0 gives $O' = a + b$. With $h = a + b + c$, we have $|O' - a| = |b| = 1$, $|O' - b| = |a| = 1$, and $|O' - h| = |-c| = 1$, so O' is the circumcenter of $\triangle HAB$. □

26 Inversion and Mixtilinear Incircles [Useful]

Theorem 26.1 (Inversion, Mixtilinear Incircles, and Casey's Theorem)**.** *Under inversion at O with radius r ($OP \cdot OP^* = r^2$):*

(a) Lines through $O \leftrightarrow$ lines through O; circles through $O \leftrightarrow$ lines not through O; circles not through $O \leftrightarrow$ circles not through O; angles are preserved; and $P^*Q^* = \frac{r^2 \cdot PQ}{OP \cdot OQ}$ ($\triangle OPQ \sim \triangle OQ^*P^*$).

*(b) **($\sqrt{bc}$-Inversion)** Inverting at A with radius $\sqrt{bc}$ and reflecting across the bisector of $\angle A$ swaps $B \leftrightarrow C$ and swaps line $BC \leftrightarrow (ABC)$.*

*(c) **(Mixtilinear Incircle)** If the A-mixtilinear incircle touches AB, AC at F, E and (ABC) internally at T, then the incenter I is the midpoint of EF (by Pascal's theorem on ABM_CTM_BC, where M_C, M_B are the arc midpoints of AB, AC).*

*(d) **(Casey's Theorem)** Four circles are tangent to a common circle Ω in cyclic order iff their external bitangents satisfy $t_{12}t_{34} + t_{23}t_{14} = t_{13}t_{24}$.*

Practice Problem 26.2. Three circles pass through O and pairwise intersect again at A, B, C. Show that under inversion at O, A^*, B^*, C^* form a triangle whose angles equal the pairwise intersection angles of the three circles at O.

Solution. Inverting at O maps the three circles through O to three lines pairwise intersecting at A^*, B^*, C^*. Since inversion preserves angles between curves, the angles between the circles at O equal the interior angles of $\triangle A^*B^*C^*$. □

27 Projective Geometry [Useful]

Definition 27.1. For collinear A, B, C, D, the **cross ratio** $(A, B; C, D) = \frac{\overline{CA}}{\overline{CB}} \cdot \frac{\overline{DB}}{\overline{DA}}$ is preserved by central projection; $(A, B; C, D) = -1$ is a **harmonic bundle**.

Theorem 27.2 (Harmonic Bundles, Poles and Polars, Brokard, Desargues, and Pascal)**.**

(a) ***(La Hire)*** *The* ***polar*** *of P with respect to $\omega(O, r)$ is the line $\perp OP$ through P^*. P lies on the polar of Q iff Q lies on the polar of P; if secant PXY meets the polar of P at Q, then $(X, Y; Q, P) = -1$.*

(b) ***(Brokard's Theorem)*** *For cyclic $ABCD$ inscribed in $\omega(O, r)$, $P = AB \cap CD, Q = AD \cap BC, R = AC \cap BD$ form a self-polar triangle $\triangle PQR$, so O is the orthocenter of $\triangle PQR$.*

(c) ***(Desargues, Pappus, Pascal, & Brianchon)*** *Two triangles are perspective from a point iff perspective from a line* ***(Desargues)****. Opposite sides of a hexagon inscribed in a conic (or pair of lines,* ***Pappus****) meet in three collinear points* ***(Pascal; proved on a circle via three applications of Menelaus and Power of a Point)****. Main diagonals of a circumscribed hexagon concur* ***(Brianchon)****.*

(d) ***(Moving Points)*** *A degree-d projective map between lines/conics is uniquely determined by $d + 1$ test positions (3 positions for a standard projectivity $d = 1$).*

Practice Problem 27.3. In $\triangle ABC$ inscribed in ω, the tangents at B, C meet at X, the tangent at A meets BC at T, and AX meets BC at K. Prove $(B, C; K, T) = -1$ and $TA^2 = TB \cdot TC$.

Solution. T lies on the polar of A (tangent at A) and the polar of X (line BC), so by La Hire the polar of T is line AX. Since the polar of T meets secant BC at K, $(B, C; K, T) = -1$. Also $\triangle TAB \sim \triangle TCA$ gives $TA^2 = TB \cdot TC$. □

28 Conic Sections and 3D Geometry [Advanced]

Theorem 28.1 (Optical Reflection Property, Cuboid Embedding, and Dandelin Spheres)**.**

(a) ***(Ellipse Reflection)*** *The tangent at P to an ellipse with foci F_1, F_2 ($PF_1 + PF_2 = 2a$) makes equal angles with PF_1 and PF_2 (every $X \neq P$ on the tangent lies outside the ellipse, so P minimizes $XF_1 + XF_2$ along the tangent line).*

(b) ***(3D Cuboid Embedding)*** *An* ***equifacial tetrahedron*** *$ABCD$ ($AB = CD = a, AC = BD = b, AD = BC = c$) inscribes in a rectangular box of dimensions x, y, z ($x^2 + y^2 = a^2$, etc.) with circumradius $R = \frac{1}{2\sqrt{2}}\sqrt{a^2 + b^2 + c^2}$ and volume $V = xyz - 4(\frac{1}{6}xyz) = \frac{1}{3}xyz$.*

(c) ***(Dandelin Spheres)*** *The two spheres tangent to a cone and to a secant plane Π touch Π at the foci F_1, F_2 of the conic section.*

Practice Problem 28.2. Prove that the product of the distances from the two foci $(\pm c, 0)$ of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ($c^2 = a^2 - b^2$) to any tangent line equals b^2.

Solution. At $P = (a\cos\theta, b\sin\theta)$, the tangent is $bx\cos\theta + ay\sin\theta - ab = 0$. The product of the distances from $(\pm c, 0)$ is

$$d_1 d_2 = \frac{|a^2b^2 - b^2c^2\cos^2\theta|}{b^2\cos^2\theta + a^2\sin^2\theta} = \frac{b^2(a^2\sin^2\theta + b^2\cos^2\theta)}{b^2\cos^2\theta + a^2\sin^2\theta} = b^2.$$ □

Part IV
Number Theory

29 Divisibility, GCD, and the Euclidean Algorithm

Definition 29.1. For integers a, b not both zero, $\gcd(a, b)$ is the largest positive integer dividing both. Integers are **coprime** if $\gcd(a, b) = 1$.

Theorem 29.2 (Division Algorithm)**.** *For integers a and $b > 0$ there are unique q, r with $a = qb + r$ and $0 \le r < b$.*

Theorem 29.3 (Euclidean Algorithm)**.** $\gcd(a, b) = \gcd(b, a - b) = \gcd(b, a \bmod b)$. *Iterating terminates and computes the gcd.*

Proof. Any common divisor of a, b divides $a - qb$ and conversely any common divisor of $b, a - qb$ divides a; so the two pairs have identical sets of common divisors, hence the same gcd. The remainders strictly decrease, so the process reaches $\gcd(g, 0) = g$. □

Theorem 29.4 (Bézout's Identity)**.** *For any a, b not both zero there exist integers x, y with $ax + by = \gcd(a, b)$. Consequently $\gcd(a, b)$ is the least positive element of $\{ax + by : x, y \in \mathbb{Z}\}$, and $\gcd(a, b) = 1$ iff $ax + by = 1$ is solvable.*

Proof. Let d be the least positive integer of the form $ax + by$. Dividing, $a = qd + r$ with $0 \le r < d$; then $r = a - qd$ is again of the form $ax' + by'$, so minimality forces $r = 0$, i.e. $d \mid a$; similarly $d \mid b$. Any common divisor of a, b divides $ax + by = d$, so $d = \gcd(a, b)$. □

Corollary 29.5 (Euclid's Lemma)**.** *If p is prime and $p \mid ab$, then $p \mid a$ or $p \mid b$. More generally if $\gcd(c, a) = 1$ and $c \mid ab$ then $c \mid b$.*

Proof. If $p \nmid a$ then $\gcd(p, a) = 1$, so $px + ay = 1$; multiply by b: $b = pbx + aby$, and p divides both terms on the right. □

Example 29.6. Compute $\gcd(2026, 748)$.

Solution. $2026 = 2 \cdot 748 + 530$; $748 = 1 \cdot 530 + 218$; $530 = 2 \cdot 218 + 94$; $218 = 2 \cdot 94 + 30$; $94 = 3 \cdot 30 + 4$; $30 = 7 \cdot 4 + 2$; $4 = 2 \cdot 2$. So gcd $= 2$. □

Practice Problem 29.7. Prove that $\gcd(2^m - 1,\ 2^n - 1) = 2^{\gcd(m,n)} - 1$.

Solution. Assume $m \ge n$. Then $2^m - 1 = 2^{m-n}(2^n - 1) + (2^{m-n} - 1)$, so $\gcd(2^m - 1, 2^n - 1) = \gcd(2^n - 1, 2^{m-n} - 1)$: the exponents perform the Euclidean algorithm, terminating at $\gcd(2^{\gcd(m,n)} - 1,\ 2^0 - 1) = 2^{\gcd(m,n)} - 1$. □

30 Primes, Unique Factorization, and Multiplicative Functions

Theorem 30.1 (Euclid)**.** *There are infinitely many primes.*

Proof. If $p_1, \dots, p_k$ were all the primes, then $N = p_1 \cdots p_k + 1 > 1$ has a prime factor, but $N \equiv 1 \pmod{p_i}$ for each i — contradiction. □

Theorem 30.2 (Fundamental Theorem of Arithmetic)**.** *Every integer $n \ge 2$ factors into primes, uniquely up to order.*

Proof. Existence follows by strong induction. For uniqueness, suppose $p_1 \cdots p_r = q_1 \cdots q_s$ are two factorizations of the smallest n with distinct factorizations. Then $p_1 \mid q_1 \cdots q_s$, so by Euclid's Lemma $p_1 \mid q_j$ for some j, forcing $p_1 = q_j$. Cancelling gives a smaller counterexample, contradiction. □

With $n = \prod p_i^{a_i}$ and $m = \prod p_i^{b_i}$:

$$\gcd(m,n) = \prod p_i^{\min(a_i,b_i)}, \quad \operatorname{lcm}(m,n) = \prod p_i^{\max(a_i,b_i)}, \quad d(n) = \prod (a_i+1), \quad \sigma(n) = \prod \frac{p_i^{a_i+1}-1}{p_i-1}.$$

30.1 Multiplicative Functions and Möbius Inversion

A function $f\colon \mathbb{N} \to \mathbb{C}$ is **multiplicative** if $f(1) = 1$ and $f(mn) = f(m)f(n)$ whenever $\gcd(m,n) = 1$. Euler's totient satisfies $\varphi(n) = n\prod_{p\mid n}\left(1-\frac{1}{p}\right)$ and $\sum_{d\mid n}\varphi(d) = n$.

Definition 30.3 (Möbius Function)**.** $\mu(n) = (-1)^k$ if n is a product of k distinct primes ($k \geq 0$), and $\mu(n) = 0$ if $p^2 \mid n$ for some prime p. It satisfies $\sum_{d\mid n}\mu(d) = [n = 1]$.

Theorem 30.4 (Möbius Inversion Formula)**.**

$$g(n) = \sum_{d\mid n} f(d) \quad \forall n \geq 1 \iff f(n) = \sum_{d\mid n} \mu(d)\, g\left(\frac{n}{d}\right) \quad \forall n \geq 1$$

Proof. $\sum_{d\mid n}\mu(d)g(n/d) = \sum_{d\mid n}\mu(d)\sum_{c\mid(n/d)} f(c) = \sum_{c\mid n} f(c)\sum_{d\mid(n/c)}\mu(d) = f(n)$. □

Practice Problem 30.5. (a) Prove that $\sqrt[k]{n}$ is irrational unless n is a perfect k-th power. (b) Prove that $\sum_{d\mid n}\varphi(d) = n$.

Solution. (a) If $\sqrt[k]{n} = a/b$, then $nb^k = a^k$, so $\nu_p(n) + k\nu_p(b) = k\nu_p(a) \implies k \mid \nu_p(n)$ for every prime p. (b) Group the n fractions $\frac{1}{n}, \frac{2}{n}, \ldots, \frac{n}{n}$ by their reduced denominator $d \mid n$ when written in lowest terms $\frac{a}{d}$ ($\gcd(a,d) = 1$); each $d \mid n$ appears $\varphi(d)$ times. □

31 Modular Arithmetic, Orders, and Primitive Roots

Theorem 31.1 (Fermat's Little Theorem, Euler, CRT, and Wilson)**.**

(a) ***(Fermat / Euler)*** *If* $\gcd(a,n) = 1$*, then* $a^{\varphi(n)} \equiv 1 \pmod n$*; for prime* $p \nmid a$*,* $a^{p-1} \equiv 1 \pmod p$ *(since* $x \mapsto ax$ *permutes the invertible residues).*

(b) ***(Chinese Remainder Theorem)*** *If* $n_1, \ldots, n_k$ *are pairwise coprime,* $x \equiv a_i \pmod{n_i}$ *has a unique solution modulo* $N = n_1 \cdots n_k$*, given by* $x = \sum_{i=1}^{k} a_i N_i M_i$ *where* $N_i = N/n_i$ *and* $N_i M_i \equiv 1 \pmod{n_i}$*.*

(c) ***(Wilson)*** $p > 1$ *is prime iff* $(p-1)! \equiv -1 \pmod p$ *(pair* $x \in \{2, \ldots, p-2\}$ *with* x^{-1}*).*

31.1 Orders and Primitive Roots

Definition 31.2. If $\gcd(a,n) = 1$, the **order** $\operatorname{ord}_n(a)$ is the least $d \geq 1$ with $a^d \equiv 1 \pmod n$. An integer g with $\operatorname{ord}_n(g) = \varphi(n)$ is a **primitive root** modulo n.

Theorem 31.3 (Fundamental Property of Order and Existence of Primitive Roots)**.**

(a) $a^m \equiv 1 \pmod n \iff \operatorname{ord}_n(a) \mid m$*. In particular* $\operatorname{ord}_n(a) \mid \varphi(n)$*.*

(b) Primitive roots modulo n exist if and only if $n = 1, 2, 4, p^k$, or $2p^k$ for an odd prime p. Every prime p has exactly $\varphi(p-1)$ primitive roots.

Proof. (a) Write $m = qd + r$ with $0 \le r < d = \mathrm{ord}_n(a)$; then $a^r \equiv a^m \equiv 1 \implies r = 0$. (b) For prime p and $d \mid p-1$, if an element a of order d exists, then $1, a, \ldots, a^{d-1}$ are all d roots of $x^d - 1 \equiv 0 \pmod p$, and the elements of order d are a^k with $\gcd(k, d) = 1$, so there are $\psi(d) = \varphi(d)$ of them. Thus $\psi(d) \le \varphi(d)$ for all $d \mid p-1$; summing over $d \mid p-1$ gives $p - 1 = \sum \psi(d) \le \sum \varphi(d) = p - 1$, forcing $\psi(d) = \varphi(d)$ for all d, including $d = p - 1$. □

Example 31.4. Find the last two digits of 7^{2026}.

Solution. Modulo 100: $7^2 = 49$, $7^4 = 2401 \equiv 1 \pmod{100}$, so $\mathrm{ord}_{100}(7) = 4$. Since $2026 \equiv 2 \pmod 4$, $7^{2026} \equiv 7^2 = 49$. □

Practice Problem 31.5. Find all positive integers n such that $n \mid 2^n - 1$.

Solution. Only $n = 1$. Suppose $n > 1$ works, and let p be the *smallest* prime divisor of n (n is odd). Then $2^n \equiv 1 \pmod p$, so $d = \mathrm{ord}_p(2)$ divides both n and $p - 1$. Since n has no prime factor $< p$, $\gcd(n, p-1) = 1$, forcing $d = 1$ and $p \mid 2^1 - 1 = 1$, contradiction. □

32 Prime Valuations, LTE, Lucas, and Cyclotomic Polynomials

Definition 32.1. For a prime p and nonzero integer x, $\nu_p(x)$ is the exponent of p in the prime factorization of x ($\nu_p(0) = +\infty$).

Theorem 32.2 (Valuations, Legendre's, Kummer's, and Lucas's Theorems)**.**

(a) $\nu_p(xy) = \nu_p(x) + \nu_p(y)$, and $\nu_p(x + y) \ge \min\{\nu_p(x), \nu_p(y)\}$ with equality *whenever $\nu_p(x) \ne \nu_p(y)$.*

(b) ***(Legendre)*** *$\nu_p(n!) = \displaystyle\sum_{i \ge 1} \left\lfloor \frac{n}{p^i} \right\rfloor = \frac{n - s_p(n)}{p - 1}$, where $s_p(n)$ is the digit sum of n in base p.*

(c) ***(Kummer)*** *$\nu_p\binom{m+n}{m} = \frac{s_p(m) + s_p(n) - s_p(m+n)}{p-1}$ equals the number of carries when adding m and n in base p.*

(d) ***(Lucas)*** *For $n = \sum n_i p^i$ and $k = \sum k_i p^i$ in base p, $\binom{n}{k} \equiv \prod_i \binom{n_i}{k_i} \pmod p$ (expand $(1+x)^n = \prod (1 + x^{p^i})^{n_i}$ in $\mathbb{F}_p[x]$).*

Theorem 32.3 (Lifting the Exponent (LTE))**.** *Let p be an odd prime, $p \mid a - b$, and $p \nmid a, b$. Then*

$$\nu_p(a^n - b^n) = \nu_p(a-b) + \nu_p(n) \qquad \big(\text{and for odd } n \text{ with } p \mid a+b:\ \nu_p(a^n + b^n) = \nu_p(a+b) + \nu_p(n)\big).$$

For $p = 2$ and odd a, b: if $4 \mid a - b$, $\nu_2(a^n - b^n) = \nu_2(a-b) + \nu_2(n)$; if n is even, $\nu_2(a^n - b^n) = \nu_2(a-b) + \nu_2(a+b) + \nu_2(n) - 1$.

Proof for odd p. Step 1 ($p \nmid n$): $a^n - b^n = (a-b)(a^{n-1} + \cdots + b^{n-1})$, and the second factor is $\equiv n a^{n-1} \not\equiv 0 \pmod p$, so $\nu_p(a^n - b^n) = \nu_p(a - b)$. *Step 2 ($n = p$):* Write $b = a + pk$. Modulo p^2,

$$\sum_{i=0}^{p-1} a^i b^{p-1-i} \equiv p\, a^{p-1} + pk\, a^{p-2} \frac{p(p-1)}{2} \equiv p\, a^{p-1} \pmod{p^2},$$

so $\nu_p(a^p - b^p) = \nu_p(a - b) + 1$. *Step 3:* Write $n = p^t m$ ($p \nmid m$), apply Step 2 t times, then Step 1. □

Theorem 32.4 (Zsigmondy, Cyclotomic Polynomials, and Wolstenholme)**.**

(a) ***(Zsigmondy)*** *For coprime $a > b \geq 1$ and $n \geq 2$, $a^n - b^n$ has a primitive prime divisor $p \equiv 1 \pmod{n}$ except when $(a, b, n) = (2, 1, 6)$ or $n = 2, a + b = 2^k$ (and $a^n + b^n$ except $2^3 + 1^3 = 9$).*

(b) ***(Cyclotomic Divisors)*** *Let $\Phi_n(x) = \prod_{\gcd(k,n)=1}(x - e^{2\pi ik/n}) \in \mathbb{Z}[x]$, so $x^n - 1 = \prod_{d|n} \Phi_d(x)$. Any prime $p \mid \Phi_n(a)$ satisfies either $p \mid n$ or $p \equiv 1 \pmod{n}$ ($\mathrm{ord}_p(a) = n$).* Proof: *If $d = \mathrm{ord}_p(a) < n$, then $x^n - 1 = (x^d - 1)\Phi_n(x)Q(x)$ has a as a double root mod p, so $p \mid na^{n-1} \implies p \mid n$.*

(c) ***(Wolstenholme)*** *For prime $p \geq 5$, the numerator of $\sum_{k=1}^{p-1} \frac{1}{k}$ is divisible by p^2 (pair $\frac{1}{k} + \frac{1}{p-k} = \frac{p}{k(p-k)} \equiv -pk^{-2} \pmod{p^2}$), and $\binom{2p-1}{p-1} \equiv 1 \pmod{p^3}$.*

Practice Problem 32.5. For odd $n \geq 1$, compute $\nu_3(2^n + 1)$, deduce $3^n \nmid 2^n + 1$ for $n \geq 2$, and show every prime $p > 3$ dividing $a^2 + a + 1$ satisfies $p \equiv 1 \pmod{3}$.

Solution. By LTE with $p = 3, a = 2, b = -1$: $\nu_3(2^n - (-1)^n) = \nu_3(2 + 1) + \nu_3(n) = 1 + \nu_3(n) < n$ for all $n \geq 2$. Also $a^2 + a + 1 = \Phi_3(a)$, so $p \mid \Phi_3(a) \implies p = 3$ or $p \equiv 1 \pmod{3}$. □

33 Diophantine Equations, Infinite Descent, and Vieta Jumping

Theorem 33.1 (Chicken McNugget, Pythagorean Triples, Pell's Equation, and Hensel)**.**

(a) ***(Chicken McNugget / Frobenius)*** *For coprime $a, b > 0$, the largest integer not of the form $ax + by$ ($x, y \geq 0$) is $ab - a - b$, and $\frac{(a-1)(b-1)}{2}$ nonnegative integers are non-representable.*

(b) ***(Pythagorean Triples)*** *Primitive solutions to $a^2 + b^2 = c^2$ ($2 \mid b$) are $a = m^2 - n^2, b = 2mn, c = m^2 + n^2$ for coprime $m > n \geq 1$ of opposite parity.*

(c) ***(Pell & Continued Fractions)*** *For non-square $d > 0$, all positive solutions to $x^2 - dy^2 = 1$ are given by $x_n + y_n\sqrt{d} = (x_1 + y_1\sqrt{d})^n$, and x_n/y_n are convergents of the continued fraction of $\sqrt{d}$.*

(d) ***(Hensel's Lemma)*** *If $P(r) \equiv 0 \pmod{p^k}$ and $p \nmid P'(r)$, there is a unique lift $r' \equiv r \pmod{p^k}$ mod p^{k+1} with $P(r') \equiv 0 \pmod{p^{k+1}}$.*

Practice Problem 33.2 (IMO 1988 Problem 6 — Vieta Jumping)**.** Let $a, b \in \mathbb{N}$ be such that $ab + 1 \mid a^2 + b^2$. Prove that $k = \frac{a^2+b^2}{ab+1}$ is a perfect square.

Solution. Suppose k is not a square. Among all nonnegative integer pairs (a, b) with $\frac{a^2+b^2}{ab+1} = k$, choose one minimizing $a + b$, with $a \geq b \geq 0$. Note $b > 0$ (else $k = a^2$). The quadratic $x^2 - kbx + (b^2 - k) = 0$ has one root a; let the other root be a'. By Vieta,

$$a + a' = kb \qquad \text{and} \qquad aa' = b^2 - k.$$

Thus $a' = kb - a \in \mathbb{Z}$, and $a' \neq 0$ since $k \neq b^2$. If $a' \leq -1$, then $a'^2 - kba' + b^2 - k \geq 1 + kb + b^2 - k > 0$, impossible. Hence $a' \geq 1$, and $a' = \frac{b^2-k}{a} < \frac{b^2}{a} \leq a \implies a' + b < a + b$, contradicting minimality of $a + b$. □

34 Dirichlet's Theorem, Sums of Squares, and Quadratic Rings

Theorem 34.1 (Dirichlet's Theorem)**.** *If* $\gcd(a,n)=1$*, the progression* $a, a+n, a+2n, \ldots$ *contains infinitely many primes.*

Lemma 34.2 (Thue's Lemma)**.** *If* $n>1$ *and* $\gcd(a,n)=1$*, there exist integers* x,y *with* $0<|x|,|y|\le\sqrt{n}$ *and* $ax\equiv y \pmod n$*.*

Proof. With $m=\lfloor\sqrt{n}\rfloor$, there are $(m+1)^2>n$ pairs $(x,y)\in\{0,\ldots,m\}^2$, so two have $ax_1-y_1\equiv ax_2-y_2 \pmod n$; set $x=x_1-x_2, y=y_1-y_2$. □

Theorem 34.3 (Two, Three, and Four Squares, and Gaussian/Eisenstein Integers)**.**

(a) ***(Fermat's Two-Square Theorem)*** *Prime* $p=x^2+y^2 \iff p=2$ *or* $p\equiv 1 \pmod 4$*;* $n=x^2+y^2 \iff 2\mid \nu_q(n)$ *for every prime* $q\equiv 3 \pmod 4$*. Proof for* $p\equiv 1 \pmod 4$*: By Wilson,* $-1\equiv\left(\left(\frac{p-1}{2}\right)!\right)^2=a^2 \pmod p$*. By Thue,* $ax\equiv y \pmod p$ *with* $0<|x|,|y|<\sqrt{p}$*, so* $p\mid x^2+y^2<2p \implies x^2+y^2=p$*.*

(b) ***(Legendre's Three-Square & Lagrange's Four-Square)*** $n=x^2+y^2+z^2 \iff n\neq 4^a(8b+7)$*; every* $n\ge 0$ *is a sum of four squares.*

(c) ***(Gaussian & Eisenstein UFDs)*** $\mathbb{Z}[i]$ *(*$N(a+bi)=a^2+b^2$*) and* $\mathbb{Z}[\omega]$ *(*$\omega=e^{2\pi i/3}, N(a+b\omega)=a^2-ab+b^2$*) are Euclidean UFDs.*

Practice Problem 34.4. (a) Prove that a prime $q\equiv 3 \pmod 4$ dividing x^2+y^2 must divide both x and y. (b) Find all integer solutions to $x^2+1=y^3$.

Solution. (a) If $q\nmid x$, then $(xy^{-1})^2\equiv -1 \pmod q \implies 1\equiv(-1)^{(q-1)/2}=-1 \pmod q$, contradiction. (b) x must be even and y odd. In $\mathbb{Z}[i]$, $(x+i)(x-i)=y^3$; since $\gcd(x+i,x-i)\mid 2i$ and $N(x+i)=y^3$ is odd, $x+i$ and $x-i$ are coprime, so $x+i=(a+bi)^3 \implies 1=b(3a^2-b^2) \implies b=-1, a=0 \implies (x,y)=(0,1)$. □

35 Quadratic Residues, Jacobi Symbol, and Reciprocity [!!]

Definition 35.1. For odd prime $p\nmid a$, the **Legendre symbol** is $\left(\frac{a}{p}\right)=+1$ if $x^2\equiv a \pmod p$ is solvable, -1 if not. For odd $n=\prod p_i^{a_i}$, the **Jacobi symbol** is $\left(\frac{a}{n}\right)=\prod\left(\frac{a}{p_i}\right)^{a_i}$.

Theorem 35.2 (Euler's Criterion and Quadratic/Jacobi Reciprocity)**.** *For odd prime* p*,* $\left(\frac{a}{p}\right)\equiv a^{(p-1)/2} \pmod p$ *and there are* $\frac{p-1}{2}$ *QRs. For coprime odd positive integers* m,n*:*

$$\left(\frac{m}{n}\right)\left(\frac{n}{m}\right)=(-1)^{\frac{m-1}{2}\cdot\frac{n-1}{2}},\qquad \left(\frac{-1}{n}\right)=(-1)^{(n-1)/2},\qquad \left(\frac{2}{n}\right)=(-1)^{(n^2-1)/8}.$$

Example 35.3. $\left(\frac{37}{103}\right)=\left(\frac{103}{37}\right)=\left(\frac{29}{37}\right)=\left(\frac{37}{29}\right)=\left(\frac{8}{29}\right)=\left(\frac{2}{29}\right)^3=(-1)^3=-1$, so $x^2\equiv 37 \pmod{103}$ has no solution.

Practice Problem 35.4. Show that for an odd prime p, every prime divisor q of 2^p-1 satisfies $q\equiv 1 \pmod{2p}$ and $q\equiv\pm 1 \pmod 8$.

Solution. $\mathrm{ord}_q(2)=p\mid q-1$ and $2\mid q-1 \implies 2p\mid q-1$. Also $2\equiv 2^{p+1}=\left(2^{(p+1)/2}\right)^2 \pmod q$, so $\left(\frac{2}{q}\right)=1 \implies q\equiv\pm 1 \pmod 8$. □

36 Prime Distribution, Approximation, and EGZ [Advanced]

Theorem 36.1 (Prime Number Theorem, Bertrand, Dirichlet Approximation, and EGZ)**.**

(a) ***(PNT & Bertrand)*** *$\pi(x) \sim \frac{x}{\ln x}$, $\prod_{p \le n} p < 4^n$, and for every $n \ge 1$ there is a prime $p \in (n, 2n]$.*

(b) ***(Dirichlet's Approximation)*** *For $\alpha \in \mathbb{R}$ and $N \ge 1$, there exist $p, q \in \mathbb{Z}$ with $1 \le q \le N$ and $|q\alpha - p| < \frac{1}{N}$ (by pigeonhole on $\{0\alpha\}, \{1\alpha\}, \ldots, \{N\alpha\} \in [0, 1)$).*

(c) ***(Erdős–Ginzburg–Ziv)*** *Among any $2n - 1$ integers, some n have sum divisible by n (multiplicative in n; for prime p, sort $0 \le a_1 \le \cdots \le a_{2p-1} < p$: either $a_i = a_{i+p-1}$ or Cauchy–Davenport on $\{a_{2p-1}\} + \sum_{i=1}^{p-1} \{a_i, a_{i+p-1}\}$ covers all p residues mod p).*

Practice Problem 36.2. Prove that $H_n = 1 + \frac{1}{2} + \cdots + \frac{1}{n}$ is never an integer for $n \ge 2$.

Solution. Let 2^k be the largest power of 2 not exceeding n ($k \ge 1$). Any other multiple of 2^k is $\ge 2^{k+1} > n$, so $m = 2^k$ is the *unique* denominator in $\{1, \ldots, n\}$ with $\nu_2(m) = k$. Hence in $H_n = \sum_{m=1}^{n} \frac{1}{m}$, exactly one term has $\nu_2 = -k$ and all others have $\nu_2 > -k$, giving $\nu_2(H_n) = -k < 0 \implies H_n \notin \mathbb{Z}$. (Alternatively, by Bertrand's postulate there is a prime $p \in (n/2, n]$, giving $\nu_p(H_n) = -1 < 0$.) □

Appendix: How to Train

- **Solve slowly, review deeply.** One problem fully understood — including *why* the key idea works and where else it applies — beats five problems rushed.
- **Keep a technique journal.** After each solved (or failed) problem, record the pivotal idea in one sentence: "smallest prime factor + order," "tangent line trick," "invert at the tangency point," "Vieta jump on minimal $a + b$."
- **Reconstruct proofs from memory.** Every theorem proved in these notes should be reproducible on a blank page; the proofs themselves are exemplars of technique (Cauchy's forward–backward induction, Kelly's extremal proof of Sylvester–Gallai, Thue's pigeonhole, Fact 5, the Menelaus proof of Pascal).
- **Sources of problems.** Past IMO, IMO Shortlists, and national olympiads; work chronologically within a topic, oldest first.
- **Mock exams.** Under real conditions: 3 problems, 4.5 hours, full write-ups. Grading your own write-up against an official solution is where most improvement happens.